Labor as Capital: AI and the Ownership of Expertise

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Abstract

Many AI tools are trained on knowledge generated by workers. When surveyed, workers report having valuable knowledge outside of their employers' current documentation. Workers also perceive substantial control over how much knowledge they share with employers. Motivated by this evidence, we develop a formal model of "knowledge supply" to AI models and show that, given current institutions and perceived control over knowledge supply, workers may restrict knowledge supply to the detriment of overall productivity. Alternative policies can recover efficiency and raise both firm and worker welfare. Individual data ownership, despite being preferred by workers, eliminates knowledge withholding but creates negative externalities for most workers: one worker's data strengthens the firm's bargaining position against others, potentially making all workers worse off. In contrast, collective data ownership achieves the first-best outcome, restoring knowledge supply while allowing workers to benefit from AI-driven productivity gains. These findings highlight the importance of labor agreements in shaping AI adoption in labor markets.

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1 Introduction

Traditionally, labor expertise resides with workers because it involves tacit knowledge developed through experience (Polanyi, 1967). Firms pay workers period by period to access skills they cannot directly possess. Yet, in modern workplaces, work increasingly generates data about work: screen recordings, communication logs, and sensor data tracking physical tasks. Once used mainly for oversight and evaluation, these data can now be repurposed to train AI systems to perform the same tasks. When this happens, workers' knowledge is transformed from labor that employers rent into capital these employers own.

Consider a common application of generative AI: coding assistants trained on software developers' code repositories. These models learn from millions of programmers' commits, pull requests, and code comments, capturing individual engineers' problem-solving approaches and stylistic conventions. Once embedded in an AI model, the expertise of individual programmers can be deployed across the organization, raising productivity for others who use the tool (Cui et al., 2025; Peng et al., 2023). The same logic applies wherever records of human labor are used to train AI models: customer service chat assistants trained on agent-customer transcripts (Brynjolfsson et al., 2025), medical treatment assistants trained on clinical knowledge (Singhal et al., 2023), or robots replicating assembly workers' movements (Zhu and Hu, 2018).

While this transformation can increase productivity, it need not benefit workers. To date, many AI systems have been trained on knowledge that workers have supplied without awareness or consent, and AI has become a central concern in modern labor disputes (Staff, 2024; Writers Guild of America West, 2023; SAG-AFTRA, 2024; Slator, 2025). At the heart of these debates is a concern about disembodiment: a worker whose core expertise has been captured by an AI system risks losing control over the source of their labor market value. This tension is not new. As factories mechanized during the Industrial Revolution, Frederick Taylor, a pioneer of scientific management, observed that informal know-how was "the principal asset or possession of every tradesman," and argued that management should surveil and codify it (Taylor, 1911). Workers resisted: restricting output, concealing efficient methods, and organizing collective actions like the Watertown Arsenal strike of 1911 (Montgomery, 1979; Aitken, 1960).

The success of AI deployment depends on access to workers' knowledge of how to perform tasks well.[1] Much of this knowledge is undocumented, residing in the experience and contextual judgment

[1] A variety of popular press accounts cite the importance of tacit or context-specific expertise. See, for instance, Murty and Kumar S (2026) and Hu (2025).

of the people who currently hold these roles. The result is a *knowledge gap*: the difference between what workers know about doing their jobs well and what their employers have captured. This gap is conceptually distinct from AI exposure indices, which measure the share of an occupation's tasks AI could in principle perform; the knowledge gap measures who holds the information required to perform them well. Closing it may require workers to supply knowledge to systems that could one day replace them.[2] In this paper, we measure the knowledge gap, show that workers perceive agency over how much knowledge they supply, and develop a formal model of knowledge supply to AI to examine the implications of alternative data governance regimes.

We measure the knowledge gap by conducting a large survey of full-time US-based workers, recruited via stratified sampling across economic sectors on Prolific. Our primary measure of the knowledge gap is the proportion of a respondent's workplace knowledge that they believe is not captured by their employer's documentation or monitoring records. To make this more concrete, we ask a conceptually related but independently framed question: how well a replacement hire, with similar ability but from another firm, would perform the respondent's job if given access only to the employer's documentation.

First, workers report a substantial gap between what they know and what their employers know. On average, workers report that roughly 40 percent of their job-relevant knowledge is not captured in their employer's documentation or records, and this gap translates into meaningful performance consequences: workers estimate that a replacement of similar ability, trained only on the employer's documentation, would perform their job less than half as well as they do. Much of the missing knowledge is tacit and situation-specific: when asked what their employer's documentation fails to capture, workers most often cite knowledge of unusual situations, of people and internal processes, and of general judgment and career skills.

We show that the knowledge gap predicts barriers to AI adoption: occupations with larger gaps show larger shortfalls in observed AI usage relative to what frontier models can theoretically perform. This is consistent with practitioner accounts that attribute low AI adoption to a lack of "context".[3] If missing knowledge is what stands between AI capability and AI deployment, that knowledge should be valuable to acquire, and we find that it is: our measured knowledge gap also

[2] In some cases, technological advances in self-reinforcement learning may preclude the need for human data.(Philip Moreira Tomei, 2026)

[3] For example, a recent Boston Consulting Group survey reports that 60% of companies report minimal revenue and cost gains from their AI investments (Boston Consulting Group, 2025), and a global IBM survey of 2,000 CEOs finds that only 25% of AI initiatives delivered expected returns (IBM Institute for Business Value, 2025). On the role of "context" specifically, see Sankar (2025). We choose to use the term "knowledge gap" (rather than "context gap") to include general occupational tacit knowledge that workers may possess, which may be independent of context.

predicts demand in the emerging market for human-generated training data, where platforms that hire experts to produce such data recruit more heavily, and spend more, in high-gap occupations.

Second, we find that the knowledge gap is not static: over 90% of workers believe they can take concrete actions to either withhold data from their employers or to supply more of it, for instance by communicating off the record or by producing more detailed documentation. In workers' own assessment, these choices have large consequences: a replacement trained on what would remain after a worker took every feasible step to withhold would perform at 45 percent of the worker's level, while one trained on the records of a fully cooperative worker would perform at 80 percent. This 35-point range implies that workers believe their supply decisions matter roughly as much as everything their employers have captured to date.

Individual agency, however, does not imply individual leverage: a worker's ability to profit from withholding depends on whether anyone else can supply the same knowledge, and we find that workers hold variable monopolies over this gap. Of the knowledge missing from their employer's documentation, workers report that colleagues could supply roughly half, while the remainder is theirs alone. This split varies widely across workers: 64 percent say their colleagues' data would be at least as valuable to a replacement as their own, while a minority report knowledge that no one else in the firm possesses. As we develop in the theory section, this variation shapes the returns to collective action.

Finally, we ask workers about their preferences over three different data governance regimes: restrictions on the use of work data for AI development, the right to bargain individually over the use of one's work data, and the right to bargain collectively over the use of work data at a firm-occupation level. We find meaningful support for all policies, with the greatest support for individual data rights.

With these empirical findings as motivation, we develop a formal model of knowledge supply from workers to AI models, and examine the welfare and distributional implications of three governance regimes: the legal status quo, in which firms own workplace data; individual data ownership; and collective bargaining over workplace data. In our model, one firm and multiple workers interact over two periods. In each period, the firm hires workers, who then decide how much knowledge to contribute. Workers differ both in their maximum potential knowledge contribution and in the cost they incur to withhold knowledge. The firm's output increases with workers' knowledge contributions and is additive across workers and across time. Workers are subject to workplace surveillance so that, through the act of working, they generate records of their processes and expertise. As a result, workers supply knowledge by default and must take costly actions to *withhold* knowledge by evading surveillance (we later allow workers to incur costs to share additional knowledge).

Knowledge supplied in period 1 becomes training data for an AI system used in period 2. Given a vector of period-1 knowledge contributions k_1, the firm trains an AI system of quality $\alpha(k_1)$, where α is a real-valued non-decreasing function. Even if a position goes unfilled, the firm can still generate output equal to $\alpha(k_1)$. AI in this model is therefore *expropriative*: the AI system improves the firm's outside option in period 2 by replicating workers' expertise. This can be interpreted either as literal AI automation or as the hiring of a low-skill worker augmented by the AI system. We assume that workers' knowledge contributions are substitutes in training the AI system, such that α exhibits decreasing differences. This captures the idea that once the firm has substantial knowledge data from some workers, the marginal contribution of additional workers' knowledge to AI capability becomes smaller.

Wages and employment in each period are determined by Nash-in-Nash bargaining between the firm and the workers. Two frictions are obstacles to efficiency. First, knowledge contributions are not contractible, so firms cannot simply offer to pay workers more in return for higher contributions. This reflects the tacit nature of much workplace expertise: firms cannot observe what individual workers know and therefore cannot write contracts that demand particular contributions. Second, period-1 contracts determine period-1 wages and employment but cannot commit parties to period-2 outcomes. As a result, workers face career concerns: their knowledge contributions today influence their bargaining position tomorrow.

In the baseline model without AI, workers have no reason to withhold knowledge. An equilibrium therefore exists in which all workers are employed in both periods, each contributes their full knowledge, and each receives a share of output proportional to their Nash bargaining weight.

Our first set of results consider the legal status quo case in which worker data is owned by the firm. Suppose, first, that workers are unaware that their data can be used to train AI models, as is likely the case for many workers today. Workers will then naively continue to supply their full knowledge in period 1. While this (weakly) increases output in period 2, it also improves the firm's outside option, weakening workers' bargaining positions and lowering their wages in period 2. The gains from AI therefore accrue primarily to the firm, raising profits while reducing worker surplus relative to the no-AI benchmark.

Workers, however, are unlikely to remain naive. Once workers anticipate how their knowledge may be used, they have incentives to preserve their bargaining position in period 2. We show that, as long as withholding costs are not prohibitive, there exists an equilibrium in which workers withhold knowledge, leading to a (weak) reduction in time-2 wages compared to the baseline no AI case. Relative to the naive case, knowledge withholding increases worker surplus, but is socially destructive, as it decreases overall (and firm) surplus. We further show that withholding is harder to sustain in equilibrium when workers are more substitutable. When colleagues can provide comparable data, an individual worker's withholding does less to protect their job from automation. Anticipating this, workers put less effort into withholding knowledge, and their period-2 wages fall as expropriation increases.

We use this model to shed light on several alternative policies: banning the use of work data for AI training, granting workers individual ownership over their work data, and establishing collective ownership of work data among workers.

Banning the use of work data for AI training, in our setting, restores the baseline no-AI equilibrium, improving workers' job security, but forgoing potential productivity gains from AI.

We model individual ownership as giving workers the right to bargain over whether their data is used at time-2, and at what price. In the event of disagreement, the firm would be unable to use that worker's data for AI model training. Individual ownership allows workers to profit from the value of their work data. As such, workers no longer have a motive to withhold knowledge, and there is an equilibrium with full knowledge contributions. Individual ownership therefore maximizes social welfare by enabling the full productivity gains from AI. Yet, for workers, this policy does not guarantee greater pay. Intuitively, individual ownership encourages each worker to contribute data, but their data has an externality: it improves the firm's bargaining position vis-a-vis other workers, lowering other workers' future wages. When workers' data are more substitutable, this effect drives down wages as the marginal value of their knowledge contributions is smaller. In the extreme case when workers are symmetric and their data are perfect substitutes, workers receive *none* of the output gains from AI, no matter their Nash bargaining weight.

The last policy we consider is one in which workers collectively own their data and bargain jointly with the firm over wages (in the event of disagreement, the firm cannot use any worker's data). Like individual ownership, collective ownership restores workers' knowledge contributions to the social first best. However, collective ownership also eliminates the competition externality present under individual ownership, because one worker's contribution no longer improves the firm's bargaining position against other workers. As a result, workers capture a larger share of the surplus generated by AI. Moreover, unlike individual ownership, collective ownership always leaves *both* workers and firms weakly better off relative to the no-AI benchmark. While our simple model abstracts from intra-union frictions, it nonetheless suggests that collective action could be a useful policy tool to ensure that workers share in the gains from AI.

Finally, motivated by our empirical evidence, we consider an extension in which workers can contribute *more* knowledge at a cost, in addition to withholding knowledge. When firms own worker data, workers have no incentives to make costly knowledge contributions, and indeed, continue strategically withholding data, as in the baseline firm-owned equilibrium. In contrast, we show that workers who individually own their data exert additional effort to codify their expertise so that they can be compensated for resulting AI improvements. However, the welfare effects of this additional effort are ambiguous: with costly contributions, it is possible that workers will exert more effort than is socially efficient, reducing total surplus compared to the no-AI case. By contrast, when workers collectively own their data, total surplus is always at least as high as in the no-AI case.

Overall, despite the popularity of individual ownership among our survey respondents, our model suggests two advantages of collective ownership. First, collective ownership always improves *worker* surplus compared to no AI, whereas individual ownership sometimes does not. Second, when workers can contribute additional knowledge at a cost, collective ownership always improves *total* surplus compared to no AI, whereas individual ownership sometimes does not.

These findings have important implications for governing the transition to AI-augmented production. While policy debates often focus on consumer data and privacy, workplace data raise distinct questions about the ownership of surveilled human capital. Greater worker awareness, absent changes in data rights, may reduce productivity as workers withhold valuable knowledge. Moreover, data rights alone may not ensure worker welfare because workers fail to internalize the negative impacts that their private data sales have on others. Our framework highlights the importance of collective action to better align workers' private incentives with social welfare.

2 Background: Worker-Enabled AI Development

2.1 Worker Data and AI Development

Recent advances in AI, particularly large language models (LLMs), rely on both public and proprietary data. Foundation models are trained on public data—internet text, books, and code—to develop general capabilities, but they often lack the domain-specific knowledge needed for particular organizational tasks.

Bridging this gap requires adapting models with proprietary data—customer support logs, call transcripts, internal documentation—often generated by workers in the course of performing these tasks. These data capture not only formal procedures but, in some cases, knowledge that was previously tacit: skills and heuristics observable in practice but difficult to codify (Polanyi, 1967).

The following examples illustrate how worker-generated data can be used to build AI systems:

> *Call Recordings and Customer Service AI*: Customer service models are trained on call transcripts and audio logs. In many cases, these transcripts are labeled not only with objective performance metrics such as call duration but also with indicators for whether the text was generated by a top-performing agent. Such labels allow the model to mimic the skills of specific individuals (Brynjolfsson et al., 2025).
>
> *Movement Video and Robotic AI*: Video of humans performing physical tasks is used to train robotic systems. For example, NVIDIA's foundation model for humanoid robots is trained in part on video of human demonstrations, and firms now pay workers to record themselves performing tasks such as folding laundry, with wearable cameras capturing their hand movements as training data (NVIDIA, 2025; Christopher, 2025).
>
> *Clinical Notes and Medical AI*: Clinical notes written by providers are used to fine-tune medical AI systems. For example, NYUTron, trained on years of clinical documentation from a single health system, is used to predict patient outcomes, generate clinical reports, and support discharge decisions (Jiang et al., 2023).

Task and workplace specific data can be captured in a variety of ways. Most notably, workplace monitoring is now pervasive and produces durable records of how labor is performed: a 2024 OECD survey finds that about 90% of U.S. firms monitor workers in some way, 75% collect data through surveillance tools, and most do not allow workers to opt out (Milanez et al., 2025; U.S. Government Accountability Office, 2024; Hertel-Fernandez, 2024). Policy attention has focused on monitoring's direct effects—privacy, safety, anxiety—with little notice that the same records can be repurposed to train AI systems that may ultimately perform the monitored tasks.[4]

In addition, workers increasingly have opportunities to be paid to supply their work knowledge. Platforms like Mercor explicitly recruit professionals such as doctors, lawyers, and financial analysts, paying over $250 per hour at times for their specialized domain knowledge (Wall Street Journal,

[4] U.S. Government Accountability Office (2024); Milanez et al. (2025); Hertel-Fernandez (2024) document concerns related to autonomy, privacy, and workplace safety, but do not mention the use of surveillance data for AI training.

2026). These markets are no longer niche: in daily scrapes of Mercor's job board between January and July 2026, we observe postings across essentially every major occupation group, including law, medicine, finance, consulting, engineering, education, and the trades, with posted rates of $50–$250 per hour and thousands of workers hired per month (see Section 3).

Finally, debate continues over whether AI will achieve human-level general intelligence (Grace et al., 2022; Allyn-Feuer and Sanders, 2023). While definitions of artificial general intelligence (AGI) emphasize broad reasoning and problem-solving capabilities (Bubeck et al., 2023), even highly capable models would still require context-specific training and examples to excel at actual workplace tasks (Mitchell, 2021). Although advanced systems may learn such knowledge autonomously, it is likely more efficient to provide models with examples from human experts who already possess this information. As a result, human-generated data and experience are likely to remain important inputs to AI systems (Ramani and Wang, 2023).

2.2 Institutional and Legal Context

The prevailing legal framework in the United States gives employers broad ownership over workplace data. Work product created within the scope of employment generally belongs to the firm under the work-for-hire doctrine, and monitoring records collected on employer systems are typically treated as firm property as well (Kim and Leavitt, 2026; U.S. Congress, 1976).

New AI capabilities are putting pressure on this framework and prompting legal and labor responses. In its 2023 contract negotiations, the Screen Actors Guild secured provisions barring studios from using recordings of an actor's likeness to create digital replicas without consent and compensation (SAG-AFTRA, 2023). More broadly, researchers and advocates have proposed governance frameworks—collective data rights, worker data trusts—intended to give workers a share in the value their data creates (Viljoen et al., 2026; Kim and Leavitt, 2026; Diamantis, 2023).

These debates point to a gap in the evidence. AI systems rely on worker-generated data, yet little is known about how much such knowledge workers hold, how willing they are to supply it, or what policies they would prefer to govern it.

3 Empirical Evidence on the Knowledge Gap

We measure the knowledge gap by conducting a large, stratified survey of full-time U.S.-based workers recruited through the survey platform Prolific. The survey collects information on participants' work, with a focus on job-specific expertise they possess, as well as their ability to influence how much of this knowledge can be recorded or documented by their employer.

3.1 Sample

We recruited 4,043 currently full-time employed U.S. workers through Prolific. The survey questions were designed around each respondent's primary job. Recruitment was stratified across economic sectors on Prolific: we set sector-level quotas so that the sample spans the full occupational distribution of U.S. full-time work.[5] Within strata, we construct post-stratification weights so that the weighted sample matches the Census distribution of full-time workers on demographics (sex, age, and education) within major (SOC 2-digit) occupation groups. Unless noted otherwise, figures and tables report weighted statistics; unweighted results are similar.

Appendix Figure A1 illustrates the breakdown of respondent occupations, with the most common being management (15.9%), business and financial operations (13.0%), office and administrative (12.1%), education and library (8.4%), and computer and mathematical (7.9%). Common job titles include manager, teacher, and software engineer. The median subject is 35 years old; 40% of subjects are male and 69% are White. We include a comparison between our sample and a cross-section of the U.S. full-time workforce in Appendix Table A1.

3.2 Measuring the Knowledge Gap

Our knowledge gap measure captures knowledge that workers possess but that their employers currently do not. We ask respondents to consider the full range of information their employer holds about their role, spanning both "official" and "unofficial" sources. Official sources are categories of data already codified and recognized as job knowledge: policy manuals, knowledge bases, training materials, and AI assistants. Unofficial sources are data the firm may be gathering but has not (yet) formally codified as job knowledge: computer activity logs, communication records, performance metrics, and work outputs. Such records are pervasive. As Appendix Figure A2 shows, a majority of respondents report that their employer monitors their performance (75%), communications (70%), outputs (63%), and time-use (56%). Substantial minorities report that they are subject to video (44%), location (44%), computer (40%), or audio recording (27%). Taking this full record as given, we ask respondents what share of their work knowledge it captures. Because we credit the employer

[5] Our resulting respondent population is more representative of U.S. workers, although it still over-represents workers in computer-based occupations because of Prolific's respondent population.

with this broad set of holdings, our definition of the knowledge gap—what workers *additionally* possess—is intended to be conservative. Finally, we also ask respondents to think about the specific aspects of their knowledge that are not well captured by their employer's documentation.

Figure 1 presents two ways of conceiving of the knowledge gap. Panel A measures it terms of *documentation*: what share of a worker's job-relevant knowledge is *not* captured by their employer's documentation or records? On average, workers report that around 40% of their workplace knowledge is not held by their employers. Split across SOC-2 occupation codes, we find that respondents in the arts, legal, and computer and mathematical fields report the largest average gaps (46–51%), while respondents in sales and transportation report the smallest (28–36%). Within-occupation variation is greater still, with an interquartile range spanning roughly 40 points in the typical occupation. Appendix Figure A3 reports the aspects of work knowledge that respondents most often say is not well-captured by their employer's documentation. The most common are knowledge of unusual situations (64%), people and internal processes (54%), and general judgment and career skills (49%), suggesting that workers hold a lot of tacit and situationally-specific knowledge.

Panel B of Figure 1 converts the knowledge gap from shares of documentation into shares of job *performance*: we ask respondents to estimate the performance of a hypothetical replacement with similar experience and general skill, trained only on the firm's documentation and records. Workers' responses indicate that they believe the documentation gaps shown in Panel A have meaningful consequences for performance: on average, workers report that such a replacement would perform their job only about half as well as they do. The occupational ranking is largely similar, with arts and legal occupations sitting at the top of both panels, and sales and transportation at the bottom. Appendix Figure A4 shows that these two measures of the knowledge gap are highly correlated at the individual level. We use the elicitation of job performance as our primary quantitative measure of the knowledge gap.

Unlike AI exposure, the knowledge gap is not only determined at the occupation level. Two workers with the same job title can report different gaps, depending on their qualifications or the circumstances of their employment. Appendix Figure A5 plots the knowledge gap against worker and firm characteristics, comparing workers within the same major occupation group. Gaps are broadly similar across education groups (Panel A) but rise with experience in the current role (Panel B) and with pay (Panel C). Firm choices matter even more: controlling for education, experience, and income, knowledge gaps are larger at firms that employ fewer monitoring technologies (Panel D).

A knowledge gap indicates what employers do not possess, but not who does. Knowledge missing from the firm's records may be shared widely among colleagues, or may reside with a single worker. Figure 2 plots, by occupation, the distribution of the share of the knowledge gap that workers believe colleagues could not fill. The average worker reports that roughly half of their uncaptured knowledge is theirs alone, though the distribution is wide, with an interquartile range from 21% to 75%. This distinction is the margin that matters for bargaining: knowledge held in common gives no individual worker leverage, while uniquely held knowledge does.

3.2.1 Knowledge Gap and AI Adoption

The knowledge gap is conceptually and empirically distinct from AI exposure. Exposure measures capture the share of an occupation's tasks that AI could technically perform (Eloundou et al., 2023); the knowledge gap measures whether the employer holds the information an AI system would need to perform them. Figure 3 plots occupation-level knowledge gaps against theoretical AI exposure at the SOC-2 level, with dashed lines at sample medians. The two are only moderately related, and occupations populate all four quadrants. Office and administrative occupations combine high exposure with a low knowledge gap: their work is both automatable and well documented, making them candidates for effective AI adoption. Computer and mathematical occupations are equally exposed but report high knowledge gaps: AI capability exists on paper, but deployment depends on capturing knowledge that firms currently do not hold. Transportation combines low exposure with a low knowledge gap, suggesting that AI adoption there awaits general technology improvements, such as robotics, rather than worker-supplied data. Food service occupations are low exposure but high knowledge gap: automation may require both technological gains and the collection of more worker data (Yang and Zhou, 2026).

This reasoning implies that where knowledge gaps are large, AI adoption should lag its technical potential. Figure 4 examines this relationship.

Using data from Massenkoff and McCrory (2026), we construct an occupation-level "Missing AI Index": the difference between an occupation's theoretical exposure to AI and its observed use, measured from Anthropic usage logs. Theoretical exposure captures the share of occupational tasks that an LLM could plausibly automate or accelerate; observed exposure reflects only those tasks that actually appear in work-related Claude interactions. Their difference measures the extent to which conceptually feasible AI use has not materialized in practice. Figure 4 shows that occupations with a high measured knowledge gap are also ones a greater gap between potential and actual AI usage: a one standard deviation increase in the knowledge gap is associated with a 0.58 standard deviation increase in the AI use gap.

If uncodified knowledge is what stands between theoretical AI capability and realized use, we should expect a market to emerge for exactly that knowledge. Figure 5 displays the positive relationship between demand for human expertise to train AI and the self-reported knowledge gap at the broad occupation level (SOC-2). In this figure, demand for human expertise is measured using daily observation of a leading job platform for AI training, Mercor, including their hourly pay rate and the number of people who have been hired for the role.[6] Occupations whose workers report larger replacement gaps see substantially more expert hiring for AI codification ($\rho = 0.66$ on log hires; 0.58 using the knowledge gap measure): the same occupations our respondents identify as running on uncaptured knowledge, such as legal, arts and media, computing, business and finance, are those where AI developers are paying to acquire it. The marketplace thus provides revealed-preference validation, from the demand side, of what our survey measures from the supply side. Appendix Figure A6 Panel A shows that demand appears first in domains with high levels of documentation, including coding and translation, and later in domains with more limited documentation including medicine, finance, and law, consistent with codification proceeding up the tacitness gradient documented above. Panel B and C that despite a weak correlation with the hourly posted rate, the overall spend in the market for expertise continues to be strongly positively correlated with our measure of the knowledge gap.

3.3 Knowledge Supply

3.3.1 Worker Actions

Having provided evidence that the knowledge gap is potentially important for productivity, we next examine workers' scope for narrowing or widening this gap.

Workers report near-universal discretion over their knowledge supply: 90% identify at least one action they could take to reduce the data their employer collects about their work, and 96% identify at least one to increase it. Figure 6 presents the share of workers reporting each action. The most common withholding options involve ordinary communication and documentation: 71% of workers could communicate off the record, and nearly half could write less detailed documentation. Fewer could withhold through more technical channels: 45% could avoid digital tracking, and roughly a

[6] 96% of postings are classified into an SOC-2 category using large language models. For details on construction, see (Cavallo and Cullen, 2026).

third could avoid physical monitoring, decline to contribute to AI training, or create fewer tutorials. Workers report even more options to supply additional data, with close to 80% able to improve their documentation or communicate more on the record.

Workers believe these actions have meaningful consequences for their replaceability. Figure 7 reports workers' beliefs about the knowledge gap facing a similarly experienced replacement, under alternative knowledge supply regimes. Recall that our baseline knowledge gap measure asked workers about the performance of a replacement worker trained only on the firm's current documentation or recordings; the average worker reported a knowledge gap of 45%. Trained instead on the documentation remaining after the individual worker took all feasible actions to reduce data collection, the gap would widen to 54%. Conversely, workers report that taking actions to supply more data can narrow the gap: if an individual worker took all feasible actions to improve documentation, then the gap would narrow to 21%. Appendix Figure A7 shows how the value of knowledge supply actions varies across occupations. Workers in every occupation believe both withholding and supplying data influence a replacement's performance. Despite sizeable cross-occupation differences in the baseline knowledge gap, workers across occupations report that supplying their individual data would allow their replacement to perform approximately 80% as well as they could perform. Workers' incentives to share knowledge therefore matter most where the baseline knowledge gap is large.

3.4 Limits to worker agency

Firms facing worker agency may respond by creating incentives for a given worker to share their expertise, but they may instead choose to limit individual worker agency by acquiring the expertise without that worker's cooperation. We focus on two versions of the second approach: greater surveillance and data from other workers.

Panel A of Figure 8 reports workers' perceptions of the maximum knowledge their employers could acquire through surveillance. Specifically, we ask them to "imagine that your employer fully monitors your work by collecting data on everything you write, say, or do at work (but not your inner thoughts or decision-making process)." Relative to what employers currently document, workers believe there is substantial scope for firms to capture knowledge through surveillance. The average worker reports that full monitoring would narrow the knowledge gap facing a replacement to around 20%, the same as what workers believe their own affirmative data contributions could achieve (Figure 7).

Firms vary in the degree to which they may be able to reduce the knowledge gap through additional surveillance. Panel B of Figure 8 shows that the returns to full monitoring are largest where current monitoring is lightest: within the same occupation and industry, workers at firms that do not monitor report nearly three times higher returns to full surveillance than workers at firms that already capture much of their data. The knowledge gap a firm faces is thus partly determined by its own choices, not just the nature of a worker or their job.

In practice, it may be difficult to fully surveil workers, or to incorporate all types of surveillance records into AI training. An alternative to relying on data contributions from a given worker is to seek contributions from other workers performing similar work. Panel A of Figure 9 presents workers' perceptions of how the knowledge gap for their own job would shrink under two scenarios: if their colleagues did everything they could to document their work (but they did not), or if they did everything they could to document their work (but their colleagues did not). On average, workers believe that colleagues' data are a good substitute for their own data, and that both can reduce the knowledge gap to around 20-25%, similar to the knowledge gap under full surveillance.

Panel B shows that the value of colleagues' data can vary across workplaces. Workers who report that more of their workplace knowledge is unique likewise perceive colleague data as less valuable (relative to their own) for reducing the knowledge gap related to their job.

Together, these results suggest that although workers report considerable agency over their own knowledge supply, firms may also have considerable scope to acquire worker knowledge without any individual worker's cooperation. As surveillance expands, passive data flows displace voluntary contributions as the primary channel of knowledge capture, and workers' ability to withhold, rather than their willingness to share, becomes the more important margin of agency. And when colleagues can supply substitute data, no worker's supply decision is independent: one worker's contribution erodes the value of another's withholding, so individual data decisions carry externalities that only coordination can internalize. Both forces raise the stakes of collective action, issues we return to in our model.

3.5 Policy Preferences

We consider worker demands for regulation of labor data, which fall into three broad categories:

> *Banning the use of AI in the workplace*: The most restrictive demand prohibits specific AI or automation uses altogether. The International Longshoremen's Association's master contract,

secured after a three-day coast-wide strike in October 2024 and ratified in February 2025, bars AI and automated systems from certain longshore functions and requires employer negotiation and worker consent before new technology is deployed (International Longshoremen's Association, 2025; USMX and ILA, 2025). In creative work, the Writers Guild of America's 2023 Minimum Basic Agreement stops short of an outright ban but reserves the Guild's right to assert that using writers' material to train AI is already prohibited under the agreement or other law (Writers Guild of America West, 2023).

Mandating individual consent: This demand accepts AI use but conditions it on the informed consent of the worker whose data is used. SAG-AFTRA's 2025 Interactive Media Agreement, ratified by 95% of voting members after an eleven-month strike, requires separate, "clear and conspicuous," reasonably specific written consent before a digital replica of a performer's voice or likeness may be created or used, and permits performers to suspend that consent during a strike (SAG-AFTRA, 2025).

Mandating collective consent: This demand moves decisions about data use from the individual to a collective structure governing pooled data under shared rules. Driver's Seat, a driver-owned cooperative, enabled ride-hail and delivery workers to pool the trip, time, and earnings data they generated across gig platforms, with members sharing in profits and electing the board (Orchi, 2023). The cooperative sold this pooled data—for instance, to local and state governments for transportation planning—so that value flowed back to the drivers who produced it (New Jersey Office of Innovation, 2026). Workers with little individual leverage over their data could thus collectively govern its use and bargain over its sale.[7]

We ask a random subset our sample of full-time US workers on their preferences over these policies related to their labor data. We present experimental vignettes to our treated subjects, describing three alternatives. The first policy bans the use of passively-collected work data in AI

[7] Driver's Seat has since been discontinued; the Workers' Algorithm Observatory at Princeton notes the project has been "sunsetted," with its mission carried forward through the FairFare tool (Workers' Algorithm Observatory, nd).

development[8]; the second allows for individual control over one's own work data, including its sale[9]; and the third allows for collective worker control and sale of work-data.[10]

To assess workers' preferences, we follow the methodology of Buckman et al. (2025).[11] Figure 10 displays the share of workers who indicate that they like each policy. While there is net positive support for all three options, individual ownership of work data emerges as the most popular choice. Approximately 65% of respondents favored a policy that would grant them the right to own and sell their individual work data for AI development.

4 Theory

4.1 Overview

We develop a model of AI expropriation to understand how worker awareness of surveillance affects knowledge sharing and welfare outcomes.

Three key features differentiate our setting from a standard employment relationship:

1. **Two periods**: today's data train tomorrow's AI, creating dynamic effects of knowledge contribution.

2. **Non-contractible knowledge contributions**: the firm does not yet know what skilled workers do until it surveils them and codifies their tacit knowledge, so it cannot write contracts specifying exactly what knowledge workers should provide.

3. **Limited commitment**: workers' expertise walks with them in the status quo—firms cannot guarantee lifetime employment, and workers cannot commit themselves indefinitely. Everything therefore happens under the shadow of renegotiation.

Our model formalizes a dynamic hold-up problem created by surveillance-enabled AI. In period 1, workers reveal tacit know-how while doing their jobs. These records train an AI system that becomes

[8]"Under this policy: 1. Employer could not use work data to develop AI models, or sell work data to other firms to develop AI models'. 2. Those seeking to develop AI models would have to hire workers to specifically produce AI training data; they could not use data that was produced as the byproduct of every day work tasks."

[9]"Under this policy: 1. You decide if your data can be used for AI training. You could say no, and your employer couldn't include your data in their AI models. 2. Your employer can pay you to use your data, at a price that you both agree on."

[10]"Under this policy: 1. You and other workers in your role jointly decide whether your collective work data can be used for AI training. You could collectively say no, and your employer couldn't include any of your data in their AI models. 2. Your employer can pay you and your colleagues to use your collective data, at a price that you all agree on. You and your coworkers could then decide how to split the proceeds amongst yourselves individually."

[11]Specifically, for each policy, we ask: "How would you feel if this policy were in place where you work?" (Positive, I would like it / Neutral / Negative, I would dislike it).

the firm's period-2 outside option. When future bargaining happens under limited commitment, a stronger outside option allows the firm to push down wages. Anticipating this, workers strategically withhold knowledge today, even though withholding is privately costly and reduces current output.

4.2 Model Setup

4.2.1 Players and Timing

The economy consists of one firm and a finite set of workers J. Time is discrete with two periods $t \in \{1, 2\}$, representing the present and the future.

Within each period

1. The firm bargains with each worker j over wage $w_t^j \geqslant 0$ and employment status $I_t^j \in \{0, 1\}$.
2. Each employed worker chooses a non-negative knowledge contribution $k_t^j \in K^j$, where K^j is a finite set.
3. Production occurs and wages are paid.

Between periods, everyone observes the vector of first-period contributions $k_1 \equiv (k_1^1, k_1^2, \ldots, k_1^{|J|})$.

We normalize each worker's output to be equal to their knowledge contribution k_t^j.[12]

We denote the vector of time-t contributions $k_t \equiv (k_t^j)_{j \in J}$, and similarly the vector of time-t wages $w_t \equiv (w_t^j)_{j \in J}$. Given some dataset $D \subseteq J$, we use k_t^D to denote the vector that is identical to k_t except that elements corresponding to $j \notin D$ are equal to 0.

We model AI quality using a function $\alpha : \mathbb{R}_{\geqslant 0}^J \to \mathbb{R}_{\geqslant 0}$. Given some time-1 contributions k_1 and some dataset D, the firm develops an AI system of quality $\alpha(k_1^D)$. Intuitively, this captures the productivity of an AI system that can augment unskilled workers or even replace workers entirely.

At time 2, for each worker j, the firm with dataset D gets output

$$\max\left\{I_2^j k_2^j, \alpha(k_1^D)\right\}. \tag{1}$$

The no-AI case corresponds to $D = \varnothing$. The functional form of (1) means that the AI automates the role, instead of augmenting the worker, because the gains from hiring worker j arise only when worker j's output exceeds what the AI could separately achieve.

[12] Our other assumptions do not rely on the cardinal properties of k_t^j, so if output is some continuous increasing function of k_t^j, we could rescale it so that each k_t^j is equal to the resulting output level.

We assume that α is monotone increasing in knowledge inputs: that is, for all $\underline{k}_1 \leqslant \overline{k}_1$, we have $\alpha(\underline{k}_1) \leqslant \alpha(\overline{k}_1)$. We assume that knowledge contributions are substitute inputs; that is, for all $\underline{k}_1^j \leqslant \overline{k}_1^j$ and all $\underline{k}_1^{-j} \leqslant \overline{k}_1^{-j}$, we have

$$\alpha(\overline{k}_1^j, \underline{k}_1^{-j}) - \alpha(\underline{k}_1^j, \underline{k}_1^{-j}) \geqslant \alpha(\overline{k}_1^j, \overline{k}_1^{-j}) - \alpha(\underline{k}_1^j, \overline{k}_1^{-j}). \tag{2}$$

Here are some examples of α that are permitted by our assumptions:

1. Linear returns $\alpha(k_1) = \sum_j \beta_j k_1^j$ for constants $\beta_j \geqslant 0$,
2. Replicating the skills of the best worker $\alpha(k_1) = \beta \max_j k_1^j$ for constant $\beta \geqslant 0$,
3. Decreasing returns to the sum of contributions $\alpha(k_1) = h\big(\sum_j k_1^j\big)$ where $h : \mathbb{R}_{\geqslant 0} \to \mathbb{R}_{\geqslant 0}$ is increasing and concave.

4.2.2 Worker Characteristics

When worker j contributes knowledge k_t^j, they incur private cost $c^j(k_t^j)$. We assume that the cost function $c^j : K^j \to \mathbb{R}_{\geqslant 0}$ has a unique minimizer $\theta^j > 0$, which attains $c^j(\theta^j) = 0$. We interpret θ^j as the worker's skill; this is the level of knowledge that they would naturally reveal when monitored, absent deliberate efforts to withhold knowledge or to record it. In general, the worker may withhold knowledge at a cost ($k_t^j < \theta^j$) or contribute more at a cost ($k_t^j > \theta^j$).

To ease exposition, we assume for now that the worker may only withhold knowledge; that is, $\theta^j = \max K^j$. Under this assumption, the costs capture strategic behavior such as sandbagging, evading surveillance, contaminating data, or degrading documentation. We will later examine how the results change with costly contributions.

4.2.3 Payoffs

We assume that workers' payoffs are additive across time, linear in wage and withholding costs, and that their outside option yields zero payoff. Thus, worker j's utility is

$$U^j = I_1^j \left(w_1^j - c^j(k_1^j) \right) + \psi \, I_2^j \left(w_2^j - c^j(k_2^j) \right), \tag{3}$$

where $\psi > 0$ is the weight on the future. We allow for the case that $\psi > 1$; one can interpret this as meaning that the stakes from expropriation are so large as to outweigh time discounting.

We assume that the firm's payoff is additive across time and across workers, and linear in output and wages. That is, the firm's utility is

$$\Pi = \sum_{j \in J} \left[I_1^j \left(k_1^j - w_1^j \right) + \psi \left(\max \left\{ I_2^j k_2^j, \alpha(k_1^D) \right\} - I_2^j w_2^j \right) \right]. \tag{4}$$

4.2.4 Solution concept

Observe that upon being hired at time 2, the unique best response for the worker is to set $k_2^j = \theta^j$, at cost $c^j(\theta^j) = 0$, and we will assume that they so do. For simplicity, we will break ties in firm-worker bargaining in favor of hiring.

An **assessment** in our model is a tuple consisting of:

1. Time-1 wages w_1.
2. Time-1 employment $\overline{J}_1$.
3. Time-1 knowledge contributions $k_1 \in \prod_{j \in J} K^j$.
4. Time-2 wages $\omega_2 : \prod_{j \in J} K^j \to \mathbb{R}^J_{\geqslant 0}$.
5. Time-2 employment $\mathcal{J}_2 : \prod_{j \in J} K^j \to 2^J$.

Note that we have specified contributions k_1^j for workers $j \notin \overline{J}_1$; these are the contributions those workers would make if (counterfactually) they were hired.

We assume that in each period, firms and workers engage in Nash-in-Nash bargaining, with exogenous bargaining weight $\gamma \in [0, 1]$ for each worker. In practice, many workplaces bargain bilaterally at the worker level (or via managers), while firm profits depend on the entire workforce. Nash-in-Nash is a tractable way to capture simultaneous bilateral bargaining while accounting for cross-worker spillovers from AI. The exogenous weight γ should be read as reduced-form bargaining strength, reflecting regulations or labor market conditions.

An assessment is an **equilibrium** if:

1. Time-1 wages w_1 and employment $\overline{J}_1$ are a Nash-in-Nash equilibrium, with the disagreement point for the worker j consisting of not being hired and not being paid.

2. For each worker $j \in J$, their contribution k_1^j maximizes their continuation payoff when the other workers contribute according to $k_1^{J_1}$.[13]

3. For each $k_1' \in \prod_{j \in J} K^j$, wages $\omega_2(k_1')$ and employment $\mathcal{J}_2(k_1')$ are a Nash-in-Nash equilibrium.

Our models differ in the firm's dataset in the event of agreement and disagreement; we describe each in the subsections that follow.

4.3 Results

4.3.1 No surveillance

Without surveillance, the firm's dataset is always $D = \varnothing$, and the disagreement point has worker j not hired and not paid.

Our model is designed to isolate the dynamic effects of expropriation by AI. Without surveillance, firms and workers are engaging in two identical and separable production stages. Knowledge contributed at time 1 has no effect on the worker's time-2 payoffs, so workers have no incentive to withhold knowledge. We state this formally in the following observation.

Observation 4.1. With no surveillance, every equilibrium has the following features:

1. full employment in both periods,

2. each worker contributes knowledge equal to their skill in both periods, $k_1^j = k_2^j = \theta^j$, and

3. wages $w_t^j = \gamma\theta^j$, where γ is the worker's Nash bargaining weight.

This equilibrium is a useful benchmark, because workers withhold no knowledge and incur no withholding costs. This equilibrium only falls short of the first-best because it achieves none of the potential productivity gains from AI.

Even under the alternative policies that follow, there will be equilibria with full employment in both periods, because the worker's outside option yields zero payoff. For ease of exposition, we will focus on these equilibria, so that the key policy-relevant comparisons we focus on are: total surplus, worker wages, and firm profits.

[13] Implicitly, this means that hired workers do not observe who else is hired when deciding their own contribution; so they best-respond to the *equilibrium* hired set $\overline{J}_1$. Notice also that workers not hired at time-1 have 'passive beliefs'. That is, if (off-path) they are hired, they believe the set of other workers hired is the same.

4.3.2 Firm-owned AI

With firm-owned AI, the firm's dataset is always $\overline{J}_1$, and the disagreement point has worker j not hired and not paid. Thus, even if no agreement is reached with worker j at time-2, the firm still produces $\alpha(k_1)$ using last period's data.

Suppose the workers naïvely contributed full knowledge at time-1. Holding fixed the workers' time-1 contributions k_1, firm-owned AI raises the output resulting from agreement between the firm and worker j at time-2, from θ^j to $\max\left\{\theta^j, \alpha\left(k_1^{\overline{J}_1}\right)\right\}$. But it also raises the firm's disagreement payoff, from 0 to $\alpha\left(k_1^{\overline{J}_1}\right)$. Thus, Nash-in-Nash bargaining results in a fall in the worker's wage, from $\gamma\theta^j$ to $\gamma\left\lfloor\theta^j - \alpha\left(k_1^{\overline{J}_1}, 0\right)\right\rfloor_+$, where we use the notation $\lfloor x\rfloor_+$ to denote $\max\{x, 0\}$. Thus, ignoring equilibrium effects, firm-owned AI increases time-2 output and the firm's time-2 profits.

Next we study what happens in equilibrium with firm-owned AI. The time-1 contribution game has a useful structure: Even though workers are harmed by raising k_1^j (it strengthens the AI they face tomorrow), the substitutes property implies the marginal harm from revealing more is smaller when others already revealed a lot. Thus, workers' time-1 contributions are strategic complements. Intuitively, if your colleagues have already "taught the AI most of what it needs," your own contribution does little incremental damage to your future wage but still saves you withholding effort $c^j(\cdot)$, so best responses are weakly increasing in others' contributions.

To ensure that our results are not vacuous, we start with a simple sufficient condition for the existence of equilibrium. We require that (essentially) withholding costs are not so severe that they outweigh a single-worker's time-1 output. Formally, let B^j be the best-response contributions of worker j when all other workers contribute 0, that is

$$B^j \equiv \arg\max_{k_1^j}\left\{w_1^j - c^j\left(k_1^j\right) + \psi\gamma\left\lfloor\theta^j - \alpha\left(k_1^j, 0\right)\right\rfloor_+\right\}. \tag{5}$$

We assume that

$$\inf\left\{k_1^j - c^j\left(k_1^j\right) : k_1^j \in B^j\right\} \geqslant 0. \tag{6}$$

We now state a result that guarantees the existence of full-employment equilibrium, and implies a (weak) reduction in time-2 wages compared to the no-surveillance case.

Theorem 4.2. With firm-owned AI, under assumption (6), there exists an equilibrium with:

1. Full employment in both periods $J = \overline{J}_1 = \mathcal{J}_2(\tilde{k}_1)$ for all $\tilde{k}_1$,

2. Time-2 wages $\omega_2^j(\tilde{k}_1) = \gamma \left\lfloor \theta_j - \alpha(\tilde{k}_1) \right\rfloor_+$ for all $\tilde{k}_1$ and all j.

Proof. We have restricted attention to equilibria with full time-2 knowledge contributions, and in every such equilibrium, Nash-in-Nash bargaining at time 2 implies that $\omega_2^j(\tilde{k}_1) = \gamma \left\lfloor \theta_j - \alpha(\tilde{k}_1) \right\rfloor_+$ for all $\tilde{k}_1$ and all j.

Given some time-1 hired set $\overline{J}_1$ and wages w_1, it follows that worker j chooses k_1^j to maximize the utility function

$$w_1^j - c^j(k_1^j) + \psi\gamma \left\lfloor \theta_j - \alpha\left(k_1^{\overline{J}_1 \cup \{j\}}\right) \right\rfloor_+ . \tag{7}$$

In order to show existence, we will establish that the simultaneous choice of k_1^j by the workers to maximize (7) is a supermodular game, in the sense of Milgrom and Roberts (1990). To do so, we first prove a technical lemma.

Lemma 4.3. Let X and Y be partially ordered sets. Suppose $f : X \times Y \to \mathbb{R}$ is monotone non-increasing[14] and has increasing differences. Suppose $g : \mathbb{R} \to \mathbb{R}$ is non-decreasing and convex. Then $g \cdot f : X \times Y \to \mathbb{R}$ has increasing differences.

We now prove Lemma 4.3. The function f has increasing differences, so for all $\underline{x} \leqslant \overline{x}$ and $\underline{y} \leqslant \overline{y}$, we have

$$f(\underline{x}, \overline{y}) - f(\overline{x}, \overline{y}) \leqslant f(\underline{x}, \underline{y}) - f(\overline{x}, \underline{y}). \tag{8}$$

By f monotone non-increasing, we have

$$\Phi \equiv f(\underline{x}, \overline{y}) - f(\overline{x}, \overline{y}) \geqslant 0, \tag{9}$$

$$f(\overline{x}, \overline{y}) \leqslant f(\overline{x}, \underline{y}). \tag{10}$$

It follows that

$$\begin{aligned} g(f(\underline{x}, \overline{y})) - g(f(\overline{x}, \overline{y})) &= g(f(\overline{x}, \overline{y}) + \Phi) - g(f(\overline{x}, \overline{y})) \\ &\leqslant g(f(\overline{x}, \underline{y}) + \Phi) - g(f(\overline{x}, \underline{y})) \text{ by (9), (10) and } g \text{ convex} \\ &\leqslant g(f(\underline{x}, \underline{y})) - g(f(\overline{x}, \underline{y})) \text{ by (8) and } g \text{ non-decreasing.} \end{aligned} \tag{11}$$

Thus, $g \cdot f$ has increasing differences. This completes the proof of Lemma 4.3.

[14] That is, for $\underline{x} \leqslant \overline{x}$ and $\underline{y} \leqslant \overline{y}$, we have $f(\underline{x}, \underline{y}) \geqslant f(\overline{x}, \overline{y})$.

Lemma 4.4. The simultaneous choice of k_1^j by the workers to maximize (7) is a supermodular game.

Most of the requirements for a supermodular game follow by inspection. The only non-trivial part is to show that for each worker j, the utility function

$$w_1^j - c^j(k_1^j) + \psi\gamma \max\left[\theta^j - \alpha\left(k_1^{\overline{J}_1 \cup \{j\}}\right)\right]_+ \tag{12}$$

has increasing differences in (k_1^j, k_1^{-j}). Observe that $\theta^j - \alpha\left(k_1^{\overline{J}_1 \cup \{j\}}\right)$ has increasing differences in (k_1^j, k_1^{-j}) by the assumption that worker contributions are substitutes, and it is monotone non-increasing. It follows that

$$\psi\gamma \max\left[\theta^j - \alpha\left(k_1^{\overline{J}_1 \cup \{j\}}\right)\right]_+ \tag{13}$$

has increasing differences by Lemma 4.3. Moreover, $w_1^j - c^j(k_1^j)$ has increasing differences trivially, because it does not depend on k_1^{-j}. The set of functions with increasing differences is closed under addition, so it follows that (12) has increasing differences in (k_1^j, k_1^{-j}). This completes the proof of Lemma 4.4.

Fixing time-1 wages w_1 and employment $\overline{J}_1$, there exists a contribution profile k_1 that satisfies the requirements of equilibrium, by Lemma 4.4 and Theorem 5 of Milgrom and Roberts (1990).

We now guess and verify that full employment at both periods, that is, $J = \overline{J}_1 = \mathcal{J}_2(\tilde{k}_1)$ for all $\tilde{k}_1$, is part of an equilibrium. This follows straightforwardly for time 2 because ω_2^j derived above gives the firm a non-negative payoff from hiring each worker.

We now consider time 1. Hiring worker j at wage w_1^j results in firm payoff

$$k_1^j - w_1^j + \sum_{l \neq j}(k_1^l - w_1^l) + \psi \sum_l \left(\alpha(k_1^J) + (1-\gamma)\left[\theta^l - \alpha(k_1^J)\right]_+\right) \tag{14}$$

and worker payoff

$$w_1^j - c^j(k_1^j) + \psi\gamma\left[\theta^l - \alpha(k_1^J)\right]_+ . \tag{15}$$

Not hiring worker j results in firm payoff

$$\sum_{l \neq j}(k_1^l - w_1^l) + \psi \sum_l \left(\alpha(k_1^{J\setminus\{j\}}) + (1-\gamma)\left[\theta^l - \alpha(k_1^{J\setminus\{j\}})\right]_+\right), \tag{16}$$

and worker payoff

$$\psi\gamma \left\lfloor \theta^j - \alpha(k_1^{J\setminus\{j\}}) \right\rfloor_+ \tag{17}$$

Thus, compared to the disagreement point, hiring worker j at time 1 increases the pairwise surplus by

$$\begin{aligned} k_1^j - c^j(k_1^j) &+ \psi\left(\max\left\{\theta^j, \alpha(k_1^J)\right\} - \max\left\{\theta^j, \alpha(k_1^{J\setminus\{j\}})\right\}\right) \\ &+ \psi \sum_{l \neq j}\left(\alpha(k_1^J) + (1-\gamma)\left\lfloor \theta^l - \alpha(k_1^J)\right\rfloor_+ - \alpha(k_1^{J\setminus\{j\}}) - (1-\gamma)\left\lfloor \theta^l - \alpha(k_1^{J\setminus\{j\}})\right\rfloor_+\right). \end{aligned} \tag{18}$$

Next we show that

$$k_1^j - c^j(k_1^j) \geqslant 0. \tag{19}$$

By hypothesis, the contribution profile k_1 is a pure-strategy Nash equilibrium of the supermodular game considered in Lemma 4.4. It follows that

$$k_1^j \in \arg\max_{\hat{k}_1^j} \left\{ w_1^j - c^j\left(\hat{k}_1^j\right) + \psi\gamma\left\lfloor \theta^j - \alpha\left(\hat{k}_1^j, k_1^{-j}\right)\right\rfloor_+ \right\}. \tag{20}$$

Let $\underline{k}_1^j$ be the smallest element of the right-hand side of (5). By Topkis' theorem (Milgrom and Shannon, 1994), we have that $\underline{k}_1^j \leqslant k_1^j$. Then by (20) and α non-decreasing we have that $c^j(\underline{k}_1^j) \geqslant c^j(k_1^j)$. By the preceding inequalities and (6), we have

$$0 \leqslant \inf\left\{\hat{k}_1^j - c^j\left(\hat{k}_1^j\right) : \hat{k}_1^j \in B^j\right\} \leqslant \underline{k}_1^j - c^j\left(\underline{k}_1^j\right) \leqslant k_1^j - c^j\left(k_1^j\right). \tag{21}$$

We have proved (19). By (19) and α monotone non-decreasing, it follows that (18) is non-negative. We have proved that full employment at time 1 is consistent with Nash-in-Nash bargaining, which completes the proof. □

We henceforth restrict attention to equilibria of the form characterized in Theorem 4.2.

Observation 4.5. With firm-owned AI, each worker's time-2 wage is lower than the no-surveillance baseline, and strictly lower if the time-1 knowledge contributions lead to positive AI quality, that is if $\alpha(k_1) > 0$.

Observation 4.6. With firm-owned AI, workers withhold knowledge at time 1 compared to the no-surveillance baseline, that is for every worker j, we have $k_1^j \leqslant \theta^j$.

Proof. With no surveillance, each worker's time-1 knowledge contribution maximizes $k_1^j - c^j(k_1^j)$, which has a unique minimizer $k_1^j = \theta^j$. With firm-owned AI, each worker's time-1 knowledge contribution maximizes (12). The observation follows by Topkis's theorem. $\square$

Observation 4.7. In equilibrium, knowledge withholding increases worker surplus and decreases firm surplus, compared to what would happen if each worker naïvely contributed $k_1^j = \theta^j$.

Proof. Let us define

$$u^j(k_1) \equiv w_1^j - c^j(k_1^j) + \psi\gamma \left\lfloor \theta_j - \alpha\left(k_1\right)\right\rfloor_+ . \tag{22}$$

Let k_1 denote an equilibrium time-1 contribution profile. We have

$$u^j(k_1^{-j}, k_1^{-j}) \geqslant u^j(\theta^j, k_1^{-j}) \geqslant u^j(\theta^j, \theta_1^{-j}), \tag{23}$$

where the first inequality is by worker best-response, and the second inequality is by u^j decreasing in the other workers' time-1 contributions. We have proved that knowledge withholding increases worker surplus. Since withholding is costly and reduces time-2 output, it reduces total surplus. It follows that it reduces firm surplus. $\square$

Under what conditions do workers withhold more knowledge in equilibrium? We now state some comparative statics results, restricting attention to equilibria of the form characterized by Theorem 4.2.

Our assumptions allow for the possibility of multiple equilibria, which can complicate comparative statics. But the workers' time-1 contributions are strategic complements, by Lemma 4.4. Thus, holding fixed the primitives, there exists a highest equilibrium, in the sense that each worker contributes weakly more at time 1 than they do in any other equilibrium (Milgrom and Roberts, 1990, Theorem 5). And there also exists a lowest equilibrium, in the sense that each worker contributes weakly less than they do in any other equilibrium. We will keep track of the highest and lowest equilibria as the primitives change.

First, we show that workers withhold less (contribute more) when withholding has a higher marginal cost.

Theorem 4.8. Consider two profiles of cost functions, $(c^j)_{j\in J}$ and $(\tilde{c}^j)_{j\in J}$, where for all workers j and contributions $k_1^j \leqslant k_1^{j\prime}$, we have

$$c^j(k_1^j) - c^j(k_1^{j\prime}) \leqslant \tilde{c}^j(k_1^j) - \tilde{c}^j(k_1^{j\prime}). \tag{24}$$

Holding all other primitives fixed, let $\overline{k}_1$ and $\underline{k}_1$ denote the highest and lowest equilibrium time-1 contributions under $(c^j)_{j\in J}$, and similarly let $\tilde{\overline{k}}_1$ and $\tilde{\underline{k}}_1$ denote the highest and lowest equilibrium time-1 contributions under $(\tilde{c}^j)_{j\in J}$. We have $\overline{k}_1 \leqslant \tilde{\overline{k}}_1$ and $\underline{k}_1 \leqslant \tilde{\underline{k}}_1$.

Proof. By Lemma 4.4, the worker payoffs (7) from time-1 contributions induced by $(c^j)_{j\in J}$ and $(\tilde{c}^j)_{j\in J}$ define a pair of supermodular games indexed by parameter τ, with $\tau = 0$ corresponding to $(c^j)_{j\in J}$ and $\tau = 1$ corresponding to $(\tilde{c}^j)_{j\in J}$. Each worker j's utility has increasing differences in (k_1^j, τ) by (24). Thus, the result follows by Theorem 6 of Milgrom and Roberts (1990). □

Next we show that equilibrium contributions rise when workers are better substitutes for each other. To state this formally, we slightly extend the model: Suppose that each worker occupies a distinct role, and let the AI quality in role j be denote $\alpha^j(k_1)$. All the results stated so far extend to this case, with essentially the same proofs.

Suppose we found a better way to use other workers' data to automate worker j's role. Intuitively, this would raise the AI quality in worker j's role for a given profile of contributions. And it would lower the marginal returns to AI quality from raising worker j's contribution. We now formally show that such a change increases equilibrium contributions.

Given two AI technologies, $\left(\alpha^j\right)_{j\in J}$ and $\left(\tilde{\alpha}^j\right)_{j\in J}$, we write that $\left(\tilde{\alpha}^j\right)_{j\in J}$ is **at least as good as** $\left(\alpha^j\right)_{j\in J}$ if for each role j, we have

$$\alpha^j(k_1) \leqslant \tilde{\alpha}^j(k_1) \text{ for all } k_1. \tag{25}$$

Given two AI technologies, $\left(\alpha^j\right)_{j\in J}$ and $\left(\tilde{\alpha}^j\right)_{j\in J}$, we write that $\left(\tilde{\alpha}^j\right)_{j\in J}$ has **stronger substitution than** $\left(\alpha^j\right)_{j\in J}$ if for each role j, we have

$$\alpha^j(\hat{k}_1^j, k_1^{-j}) - \alpha^j(k_1^j, k_1^{-j}) \geqslant \tilde{\alpha}^j(\hat{k}_1^j, k_1^{-j}) - \tilde{\alpha}^j(k_1^j, k_1^{-j}) \text{ for all } k_1^j \leqslant \hat{k}_1^j \text{ and all } k_1^{-j}. \tag{26}$$

Theorem 4.9. Suppose that $\left(\tilde{\alpha}^j\right)_{j\in J}$ is at least as good as $\left(\alpha^j\right)_{j\in J}$ and that $\left(\tilde{\alpha}^j\right)_{j\in J}$ has stronger substitution than $\left(\alpha^j\right)_{j\in J}$. Holding all other primitives fixed, let $\overline{k}_1$ and $\underline{k}_1$ denote the highest and lowest equilibrium time-1 contributions under $\left(\alpha^j\right)_{j\in J}$, and similarly let $\tilde{\overline{k}}_1$ and $\tilde{\underline{k}}_1$ denote the highest and lowest equilibrium time-1 contributions under $\left(\tilde{\alpha}^j\right)_{j\in J}$. We have $\overline{k}_1 \leqslant \tilde{\overline{k}}_1$ and $\underline{k}_1 \leqslant \tilde{\underline{k}}_1$.

Moreover, worker wages at time-2 are lower under technology α and contribution profiles $\tilde{\overline{k}}_1$ and $\tilde{\underline{k}}_1$, compared to technology $\tilde{\alpha}$ and contribution profiles $\overline{k}_1$ and $\underline{k}_1$ respectively.

The proof is in Section B.1.

4.4 Alternative Policies

4.4.1 Individual data ownership

Under individual-owned AI, the firm's dataset D is equal to the set of workers hired at *both* time-1 and time-2. Thus, the disagreement point for bargaining at time-2 between the firm and worker j involves the worker not being hired, not being paid, and their data being omitted from the dataset.

Since we are breaking ties in favor of employment and the pairwise surplus from employing an additional worker is at least zero, every equilibrium with individual-owned AI involves full employment at time 2. In such an equilibrium, the pairwise surplus from employing worker j at time-2 is

$$\lambda_j(k_1, \overline{J}_1) \equiv \max\left\{\theta^j, \alpha\left(k_1^{\overline{J}_1}\right)\right\} - \alpha\left(k_1^{\overline{J}_1\setminus\{j\}}\right) + \sum_{l\neq j}\left(\max\left\{\theta^l, \alpha\left(k_1^{\overline{J}_1}\right)\right\} - \max\left\{\theta^l, \alpha\left(k_1^{\overline{J}_1\setminus\{j\}}\right)\right\}\right) \tag{27}$$

Observe that $\lambda_j(k_1, \overline{J}_1)$ is non-decreasing in k_1^j, by α monotone non-decreasing. By Nash-in-Nash bargaining, worker j's equilibrium wage is $\gamma\lambda_j(k_1, \overline{J}_1)$. At time 1, worker j chooses k_1^j to maximize

$$w_1^j - c^j(k_1^j) + \psi\gamma\lambda_j(k_1, \overline{J}_1), \tag{28}$$

and by c^j non-increasing and λ_j non-decreasing in k_1^j, it follows that it is a best response to contribute $k_1^j = \theta^j$. Thus, the pairwise surplus from employing any worker j at time 1 is at least zero, so there is an equilibrium with full employment at time 1.

Observation 4.10. Under individual-owned AI, there is an equilibrium with full employment in both periods and with $k_1^j = k_2^j = \theta^j$.

Henceforth we restrict attention to the equilibrium described in Observation 4.10. Notice that, compared to firm-owned AI, workers contribute weakly more knowledge at time 1, which raises output directly at time 1 directly and at time 2 indirectly, by improving AI quality.

Observation 4.11. Total surplus is higher under individual ownership than under firm ownership.

However, individual ownership does not guarantee that workers are better off, compared to the no-AI baseline. The intuition for this is that Nash-in-Nash bargaining compensates workers based on the marginal contribution of their data, and under substitutes the total contribution exceeds the sum of the marginal contributions. That is,

$$\alpha(k_1^J) - \alpha(0) \geqslant \sum_j \left(\alpha(k_1^J) - \alpha(k_1^{J\setminus\{j\}})\right). \tag{29}$$

To see this, observe that we can number the workers arbitrarily from 1 to $|J|$, and rewrite the left-hand side of (29) as a telescoping sum

$$\alpha(k_1^J) - \alpha(0) = \sum_{l=1}^{|J|} \left(\alpha(k_1^{\{j:j\leqslant l\}}) - \alpha(k_1^{\{j:j\leqslant l-1\}})\right) \geqslant \sum_j \left(\alpha(k_1^J) - \alpha(k_1^{J\setminus\{j\}})\right), \tag{30}$$

where the inequality follows by the substitutes assumption.

Individual ownership generates a competition externality: Each worker accounts for the (non-negative) effect of their time-1 contribution on their time-2 wage, but does not account for how their time-1 contribution decreases other workers' time-2 wage. Thus, individual ownership does not ensure that workers receive a substantial share of the rents from AI. This problem is worse when workers' data are more substitutable for each other, as we now state formally.

Theorem 4.12. Suppose that $\left(\tilde{\alpha}^j\right)_{j\in J}$ has stronger substitution than $\left(\alpha^j\right)_{j\in J}$ and also that

$$\tilde{\alpha}^j(\theta) = \alpha^j(\theta) \text{ for all } j. \tag{31}$$

Then each worker's equilibrium payoff is lower under $\left(\tilde{\alpha}^j\right)_{j\in J}$ than under $\left(\alpha^j\right)_{j\in J}$.

The proof is in Section B.2.

In some cases, individual ownership can harm workers compared to the no-AI baseline, even when workers have full Nash bargaining power. We now state this formally.

Theorem 4.13. Suppose that:

1. There are at least three workers,
2. workers have identical maximum contributions, with $\theta^j = \theta^{j'}$ for all j and j',

3. contributions are perfect substitutes, that is $\alpha(k_1) = f(\max_j\{k_1^j\})$ for some increasing function f.

Then for any Nash bargaining parameter $\gamma \in (0, 1]$, the full-contribution full-employment equilibrium under individual data ownership is strictly worse for each worker than the full-contribution full-employment equilibrium under no surveillance.

The proof is in Section B.3.

To summarize, individual data ownership restores efficiency, because it ensures that each worker has no incentive to withhold knowledge at time-1. But individual data ownership does not guarantee that workers share in the efficiency gains from AI. By Theorem 4.13, under some conditions workers can be strictly worse off under individual data ownership, compared to the no-surveillance case. Under those conditions, firms benefit from AI not only because it raises time-2 output, but also because it suppresses workers' time-2 wages.

4.4.2 Collective data ownership

So far we have considered workers bargaining individually with the firm, via Nash-in-Nash bargaining. Suppose instead that the workers bargain as a single union with the firm. The union has the same Nash bargaining weight γ, and a utility equal to the sum of the individual worker utilities. Suppose furthermore that in the event of disagreement at time 2, no workers are employed and the firms dataset is $D = \varnothing$. That is, we will call an assessment an **equilibrium with collective ownership** if:

1. Time-1 wages w_1 and employment $\overline{J}_1$ are a Nash bargaining solution between the firm and the union.

2. For each worker $j \in J$, their contribution k_1^j maximizes their continuation payoff when the other workers contribute according to $k_1^{J_1}$.

3. For each $k_1' \in \prod_{j \in J} K^j$, wages $\omega_2(k_1')$ and employment $\mathcal{J}_2(k_1')$ are a Nash bargaining solution between the firm and the union.

Observe that so long as worker j's wage $\omega_2^j(k_1)$ is non-decreasing in k_1^j, it is a best response for worker j to contribute $k_1^j = \theta^j$ at time 1. One example of such a scheme is to pay each worker an equal share $\gamma/|J|$ of output in each period.

Observation 4.14. There is an equilibrium with collective ownership with full employment in both periods and full knowledge contributions $k_t^j = \theta^j$ in both periods.

Under collective ownership, the time-2 disagreement point excludes *all* workers' data. This bundles individual data rights so that when one worker contributes, they do not inadvertently strengthen the firm's hand against others. This prevents competition externalities from arising, ensuring that workers share in the gains from AI. Next, we state that the benefits to workers of collective ownership are larger when their data are better substitutes for each other.

Observation 4.15. Suppose that $\left(\tilde{\alpha}^j\right)_{j\in J}$ has stronger substitution than $\left(\alpha^j\right)_{j\in J}$ and also that $\tilde{\alpha}^j(\theta) = \alpha^j(\theta)$ for all j. Then the gains to worker surplus of switching from individual ownership to collective ownership are larger under $\left(\tilde{\alpha}^j\right)_{j\in J}$ than under $\left(\alpha^j\right)_{j\in J}$.

Proof. Observe that under collective ownership, worker surplus under $\left(\alpha^j\right)_{j\in J}$ is

$$\gamma \sum_j \left(\theta^j + \psi \max\left\{\theta^j, \alpha^j(\theta)\right\}\right), \tag{32}$$

and worker surplus under $\left(\tilde{\alpha}^j\right)_{j\in J}$ is

$$\gamma \sum_j \left(\theta^j + \psi \max\left\{\theta^j, \tilde{\alpha}^j(\theta)\right\}\right). \tag{33}$$

By $\tilde{\alpha}^j(\theta) = \alpha^j(\theta)$, these are equal. The observation then follows by Theorem 4.12. □

4.5 Extension to costly contributions

So far we have assumed that the worker may only withhold knowledge at a cost; that is, $\theta^j = \max K^j$. We now consider the more general model, that permits $\theta^j < \max K^j$. Recall that θ^j is defined as the unique minimizer of the worker's cost function, which attains $c^j(\theta^k) = 0$. Moreover, this is the worker's equilibrium contribution in the no-surveillance baseline. Thus, by permitting $\theta^j < \max K^j$, we are positing that the worker may contribute strictly more knowledge than that baseline, at some personal cost.

Our analysis of the no-surveillance baseline and of firm-owned AI extends without modification to the case with costly contributions. The results are true as stated, and the proofs do not rely on the assumption that $\theta^j = \max K^j$.[15]

[15] That is, the following results extend with no modification: Observation 4.1, Theorem 4.2, Observation 4.5, Observation 4.6, Observation 4.7, Theorem 4.8, and Theorem 4.9.

Under individual data ownership, allowing costly contributions changes the analysis technically and substantively. When $\theta^j = \max K^j$, the worker's time-1 best response is to contribute $k_1^j = \theta^j$ regardless of the other workers' contributions, because their time-2 wage is weakly increasing in k_1^j and their time-1 costs are strictly decreasing in k_1^j. By contrast, if $\theta^j < \max K^j$, then the worker's time-1 best response may depend on the other workers' contributions.

Consequently, the technical change to the analysis is that there may not be an equilibrium in pure strategies for the workers.[16] Nonetheless, if we extend the definition of equilibrium to allow workers to mix over time-1 knowledge contributions, then an equilibrium exists. Since each worker's set of possible contributions K^j is finite, and the continuation payoffs resulting from each contribution are well-defined (as in the proof of Theorem 4.2), this follows by the existence of Nash equilibrium for finite games. Under a technical assumption, such an equilibrium also has full employment in both periods.

Theorem 4.16. Suppose that for every worker j, we have

$$\inf_{k_1^j \in K^j} \left\{ k_1^j - c^j(k_1^j) \right\} \geqslant 0. \tag{34}$$

Under individual-owned AI, there is an equilibrium (possibly with randomization in time-1 contributions) with full employment in both periods.

The proof is in Section B.4.

The substantive change to the analysis is that, with costly contributions, equilibrium contributions can be strictly higher than in the no-surveillance baseline. This is because workers are rewarded for their marginal contribution to AI quality, and can increase that contribution at a cost.

Recall that under firm ownership, equilibrium contributions never exceed θ^j (Observation 4.6). The conclusion that individual ownership raises contributions compared to firm ownership holds even under costly contributions, as we now state formally.

Theorem 4.17. Under any equilibrium with individual ownership, if worker j is hired and contributes k_1^j with positive probability, then $k_1^j \geqslant \theta^j$.

The proof is in Section B.5.

However, the welfare effects of individual ownership can change under costly contributions; Observation 4.11 does not follow without further assumptions. The key is that under individual

[16] This complication does not arise under firm ownership of data because worker contributions at time-1 are a supermodular game; see Lemma 4.4.

ownership, workers may make costly contributions that are individually rational but not socially beneficial, as in the following example.

Example 1. There are two workers, with $K^1 = K^2 = \{5, 6\}$. The AI has quality 3 if one worker chooses contribution 6, quality 4 if both workers choose contribution 6, and 0 otherwise. Contribution 5 is free (so $\theta^1 = \theta^2 = 5$), and contribution 6 costs $\epsilon > 0$. In the unique equilibrium under firm ownership both workers contribute 5 at time 1. For ϵ small enough, there exists an equilibrium under individual ownership in which both workers contribute 6 at time 1. In the latter equilibrium, both workers pay positive costs but total output has not increased, so total surplus is strictly lower than under firm ownership, and also strictly lower than in the no-AI case.

Next we study an equilibrium with collective ownership, assuming for simplicity that the union splits total compensation in each period equally across workers. In general, the comparison of total surplus between individual ownership and collective ownership is ambiguous. Example 1 proves that individual ownership can sometimes result in strictly lower total surplus than the no-AI case. By contrast, collective ownership always raises total surplus compared to the no-AI case, as we now state formally.

Theorem 4.18. Under any equilibrium with collective ownership, expected total surplus is at least as high as in the no-AI case.

The proof is in Section B.6.

5 Conclusion

AI models can now perform a wide range of workplace tasks. But in many cases, model performance depends on access to detailed knowledge about how work is actually done, knowledge that has traditionally belonged to workers. While many foundational AI models were trained on historical records collected without workers' awareness or meaningful consent, further progress will depend on new data from workers who are increasingly aware that their actions, communications, and outputs can be used to train systems that replicate their work.

Our survey of U.S. full-time workers suggests that, even with growing workplace surveillance, there still remains a substantial knowledge gap between what workers know about how do their jobs well, and what their employers have documented. Training AI systems may require firms to find ways to encourage workers to share their additional expertise.

We provide evidence that workers' have agency in choosing how much knowledge to supply to their employers. This empirical finding motivates our theory, which considers one important reason why workers may want to adjust their knowledge supply: career concerns. We show that, under current institutional arrangements, concerns about expropriation give workers an incentive to hide knowledge. Such behavior can have negative consequences for all parties: workers can suffer lower productivity and wages in the present, firms can face lower profits from reduced ability to make use of labor expertise, and overall economic output can decline due to lower productivity gains from the adoption of effective AI systems. These frictions give workers, employers, and policymakers a shared interest in governance structures that mitigate fears of data-driven expropriation.

Our analysis also reveals a tension between workers' stated preferences and their longer-term welfare. Roughly 65% of respondents favor individual control over their work data, including the right to sell it for AI development. Our theory shows, however, that unilateral sales create a competition externality: a given worker's decision to sell their data improves the firm's outside option against all other workers. Because workers do not internalize this spillover, individual data sales can leave all workers worse off relative to a no-AI benchmark, even when they hold full bargaining power. By contrast, collective data ownership internalizes these externalities and increases worker surplus relative to no AI. Implementing such arrangements, however, raises practical challenges. Differences in the value of workers' contributions may generate intra-union tensions, and the legal basis for worker ownership is difficult to define when data are produced using firm-provided capital and intellectual property.

More broadly, our results suggest that the governance of workplace data will play a central role in shaping the trajectory of AI adoption and worker welfare. If current arrangements persist, rising worker awareness may slow the development of effective AI by inducing meaningful resistance, as evidenced by growing labor disputes around AI in the workplace, or by merely failing to motivate valuable knowledge contributions. Designing, testing, and refining alternative mechanisms, whether through public policies or private contracts, is an important direction for future research.

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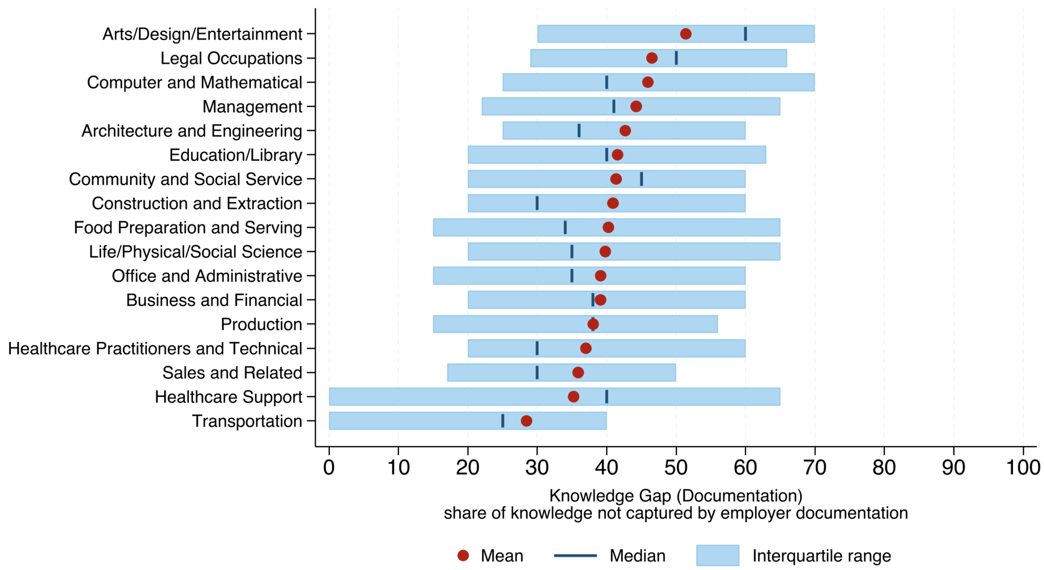
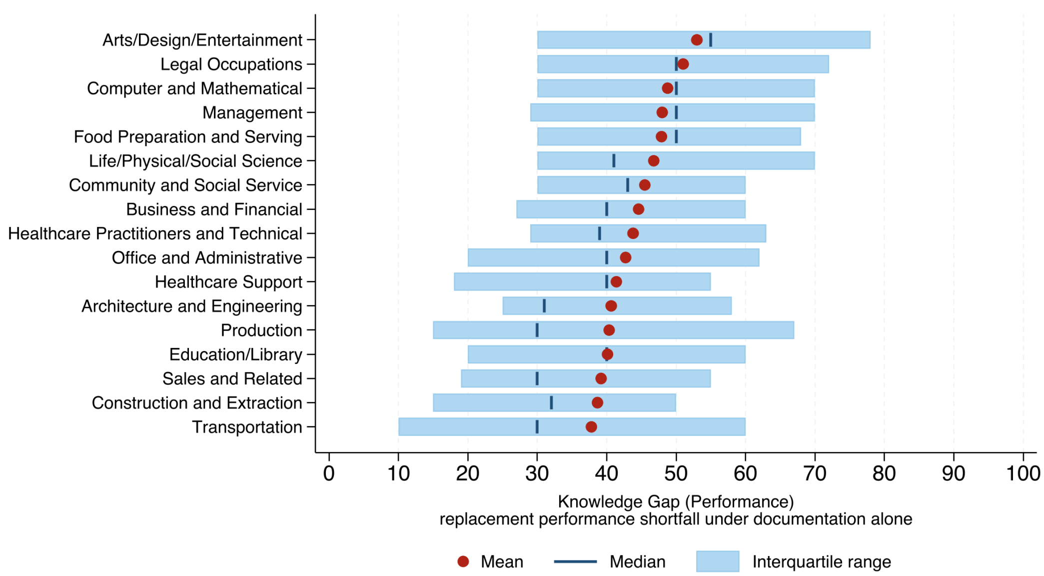
Figure 1: Distribution of Surveyed Knowledge Gap, by Occupation

A. Documentation Shortfall

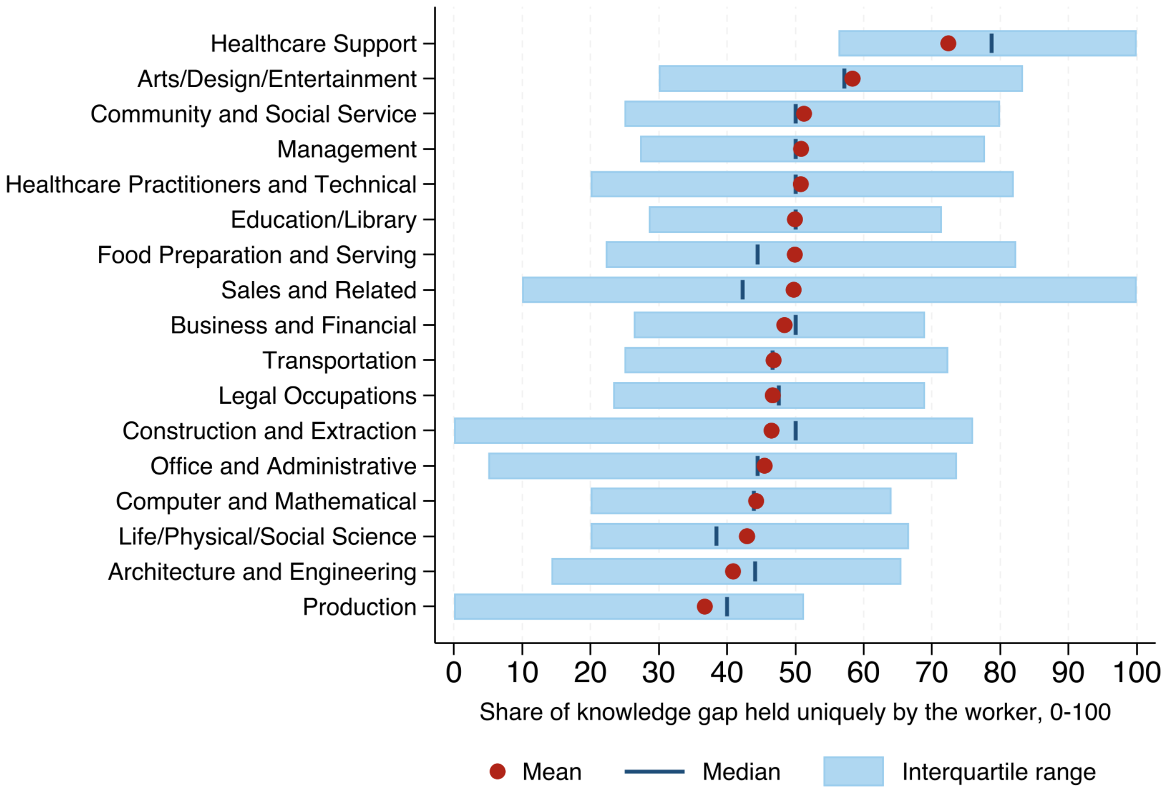


B. Replacement Performance Shortfall



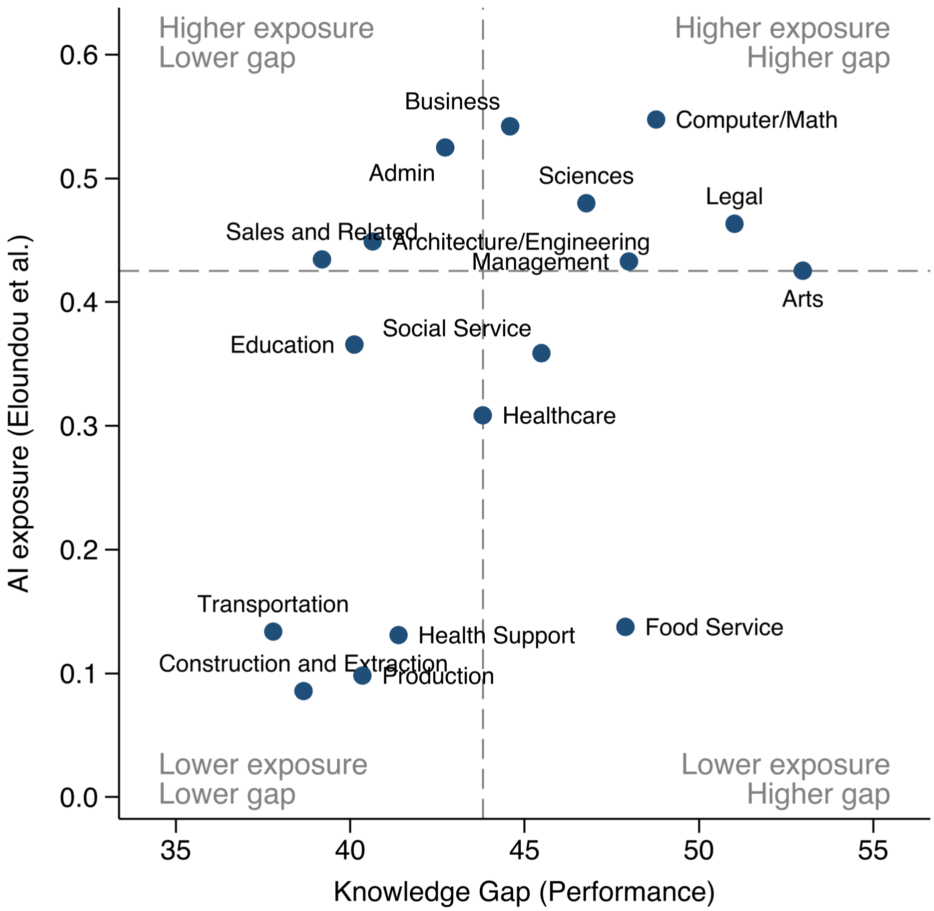
Notes: Both panels are by SOC 2-digit occupation group, sorted by the group mean; red dots mark occupation means, vertical ticks medians, and shaded bands interquartile ranges. Occupation groups with fewer than 40 respondents are omitted. Panel A shows the distribution of the knowledge gap: 100 − the response to "*On a scale of 0 to 100, how much of your overall work knowledge is captured by your employer's documentation?*" Panel B shows the distribution of the replacement gap: 100 − the response to "*Remember that your replacement has similar experience to you (same education, ability, and experience, but from another firm). Your replacement is able to learn ONLY from your employer's documentation. On a scale of 0 to 100, how well would your replacement be able to do your job?*"

Figure 2: Worker's Unique Share of the Knowledge Gap, by Occupation



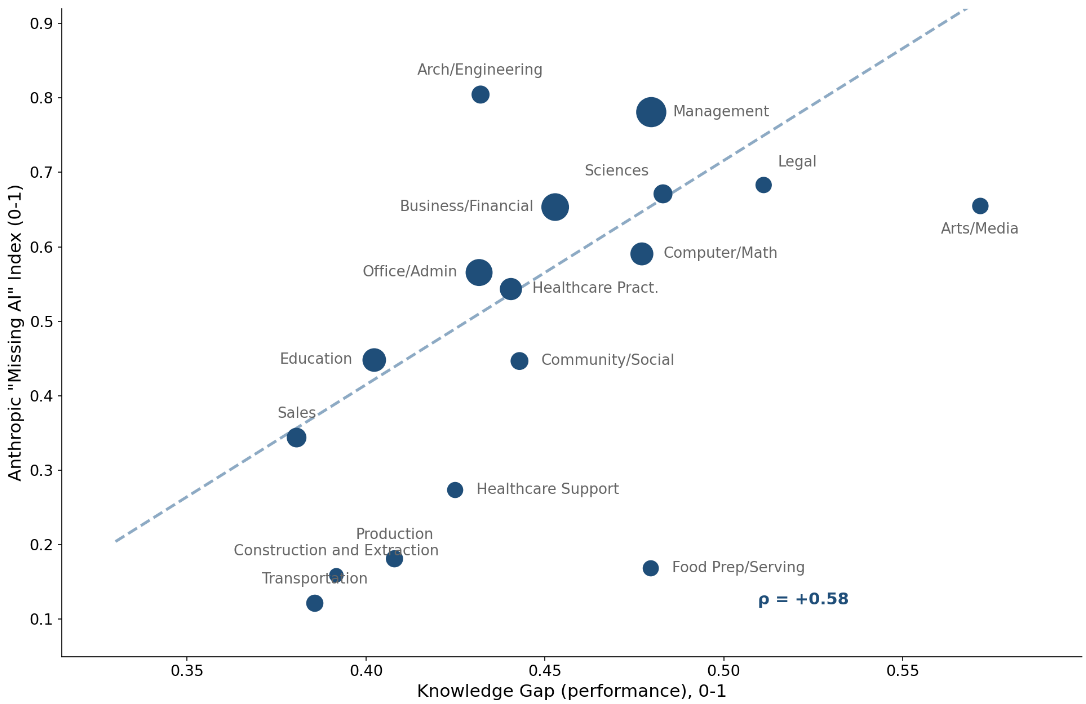
Notes: The figure shows the distribution of the worker's unique share of the knowledge gap by SOC 2-digit occupation group: the share of the knowledge gap that colleagues cannot fill, i.e., knowledge captured by neither employer documentation nor colleagues, as a share of all knowledge not captured in employer documentation. Groups are sorted by the group mean; red dots mark occupation means, vertical ticks medians, and shaded bands interquartile ranges. Occupation groups with fewer than 40 respondents are omitted. The sample covers respondents with a positive documentation gap and internally consistent responses (documentation-or-colleague coverage at least documentation coverage).

Figure 3: Knowledge Gap vs. AI Exposure



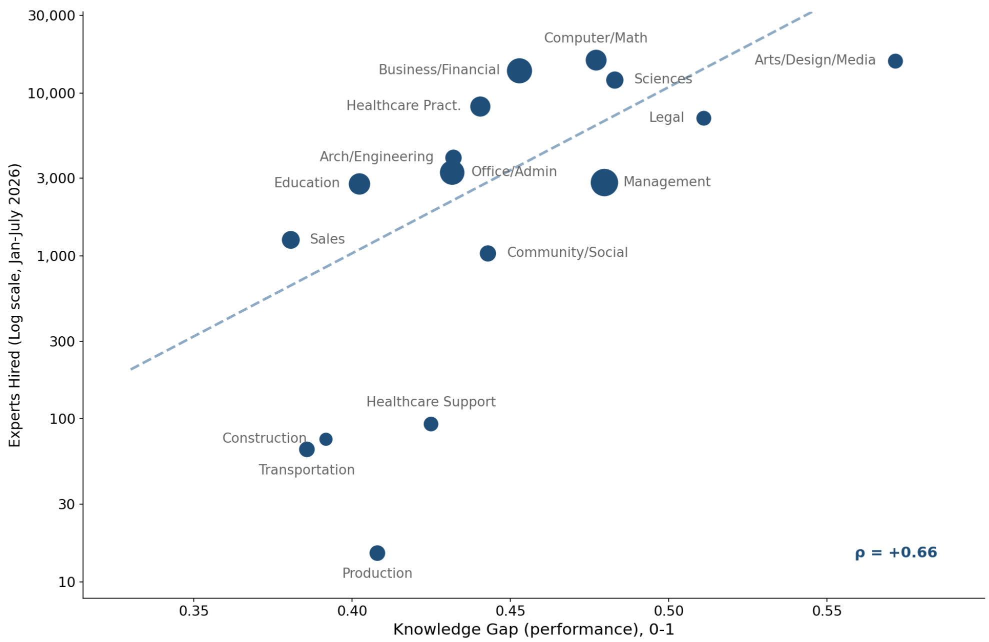
<u>Notes:</u> Each point is a SOC 2-digit occupation group with at least 40 survey respondents (17 groups). The horizontal axis is the occupation's mean Knowledge Gap (Performance): the self-reported shortfall in a similar replacement's performance when trained only on employer documentation, i.e., 100 minus the response to "*Your replacement is able to learn ONLY from your employer's documentation. On a scale of 0 to 100, how well would your replacement be able to do your job?*", averaged within occupation (0–100 scale). The vertical axis is the occupation's theoretical AI exposure developed by Eloundou et al. (2023), measuring the share of the occupation's work tasks exposed to LLMs; we use the human-rated series, weight tasks by O*NET core importance, and aggregate SOC 6-digit exposure to the SOC 2-digit level using 2025 OEWS employment weights. Both axes are in levels; dashed lines mark the unweighted medians of each measure across the 17 occupation groups, and the corner labels describe the resulting quadrants.

Figure 4: Knowledge Gap and "Missing" AI Adoption



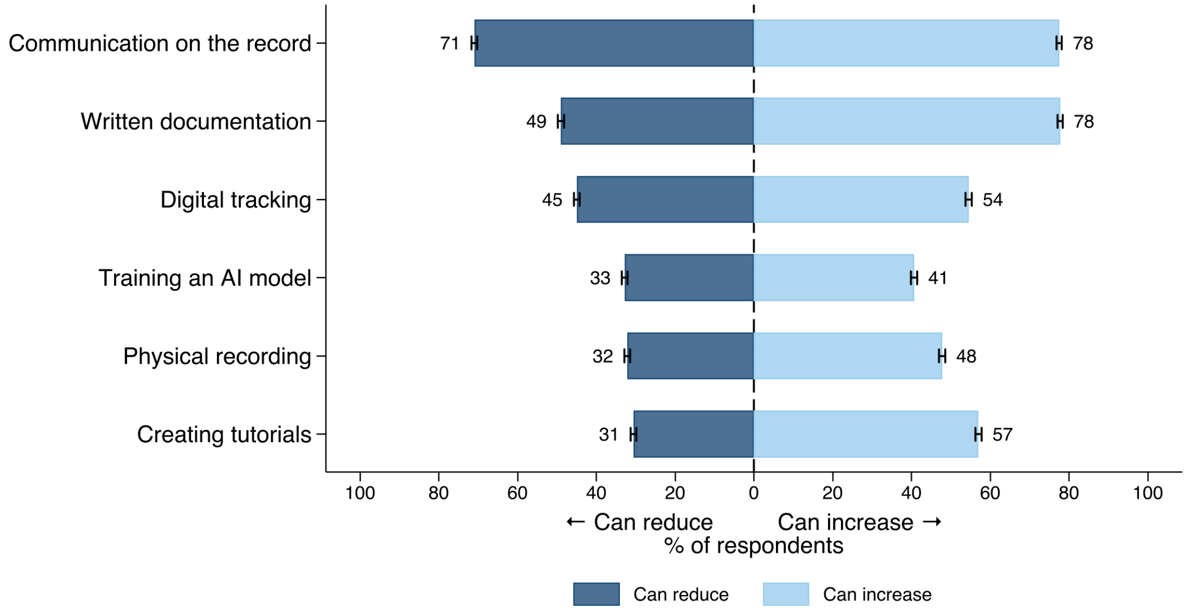
<u>Notes:</u> Each point is a SOC 2-digit occupation group; marker size is proportional to the number of survey respondents in the group, the dashed line is the OLS fit, and ρ reports the correlation. Survey gaps are computed at the individual level and collapsed to the SOC 2-digit occupation mean; only occupation groups with at least 40 respondents are retained. The figure shows the Anthropic "Missing AI" Index scattered against our survey's replacement gap measure. The replacement gap is the self-reported shortfall in a similar replacement's performance when trained only on employer documentation: 100 minus the response to the question "*Your replacement is able to learn ONLY from your employer's documentation. On a scale of 0 to 100, how well would your replacement be able to do your job?*", rescaled to 0–1. The "Missing AI" Index, taken from Massenkoff and McCrory (2026), is theoretical AI use minus observed AI use, so higher values mean that more of the conceptually feasible AI use in an occupation has not yet materialized in practice. Correlations are suggestive given the small number of occupation groups.

Figure 5: Knowledge Gap and the Market for Human Expertise



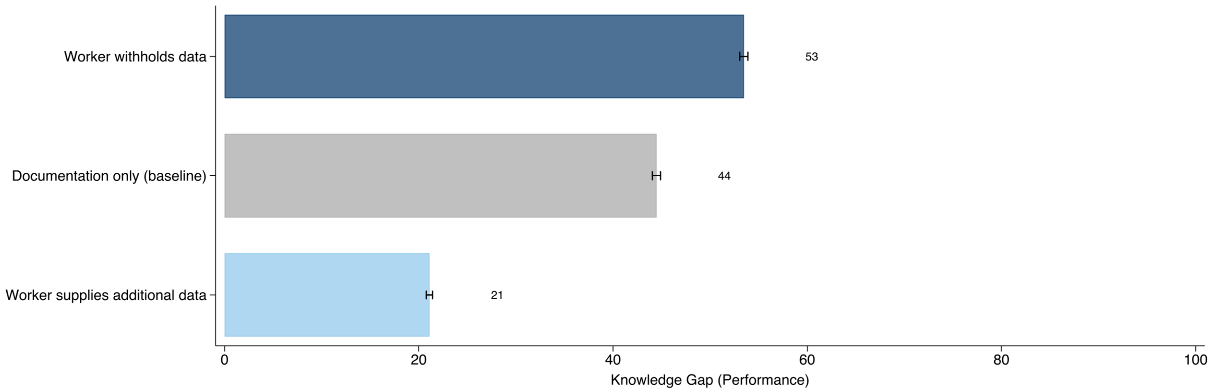
Notes: The figure plots the number of workers hired through Mercor between January 20 and July 13, 2026 (log scale) against the occupation group's replacement gap (defined in Figure 4). Mercor is a leading marketplace through which AI developers hire domain experts to supply training data. Each point is one of the 16 SOC 2-digit occupation groups with Mercor hiring; marker size is proportional to the number of survey respondents in the group, the dashed line is the OLS fit (weighted by survey respondents), and ρ reports the correlation, computed on $\log_{10}$ hires. Hires are measured from daily observations of Mercor's public job board: each posting is classified from its title to a detailed SOC occupation (96% of postings assigned to a SOC 2-digit group). Because the posted "N hired this month" counter reflects a trailing 30-day window rather than a calendar-month count, a posting's hires are its counter level when first observed plus all subsequent day-over-day increases; occupation totals sum over postings. Appendix Figure A6 presents the Knowledge Gap's relationship to timing, posted prices, and spend.

Figure 6: Actions Workers Can Take to Adjust Knowledge Supply



Notes: Respondents answered two mirrored multi-select questions asking which actions they could take to "*REDUCE*" or "*INCREASE*" "*the amount of data your employer is able to collect about your work activities*" (the increase version adds "*or your thought process behind those activities*"). Dark blue bars give the share selecting an action in the reduce question; light blue bars, the corresponding action in the increase question. Rows: communication on the record (e.g., using email or work chat rather than speaking off the record); written documentation (e.g., comments, saved drafts, explanations of choices); digital tracking (e.g., screen recordings, activity logs); physical recording (e.g., video or audio of work activities); training an AI model (e.g., rating or correcting AI outputs, sharing documents to improve its knowledge); and creating tutorials (recording-based documentation such as video tutorials or explainers). Rows are sorted by the share who could reduce data collection. Whiskers span ±1 standard error.

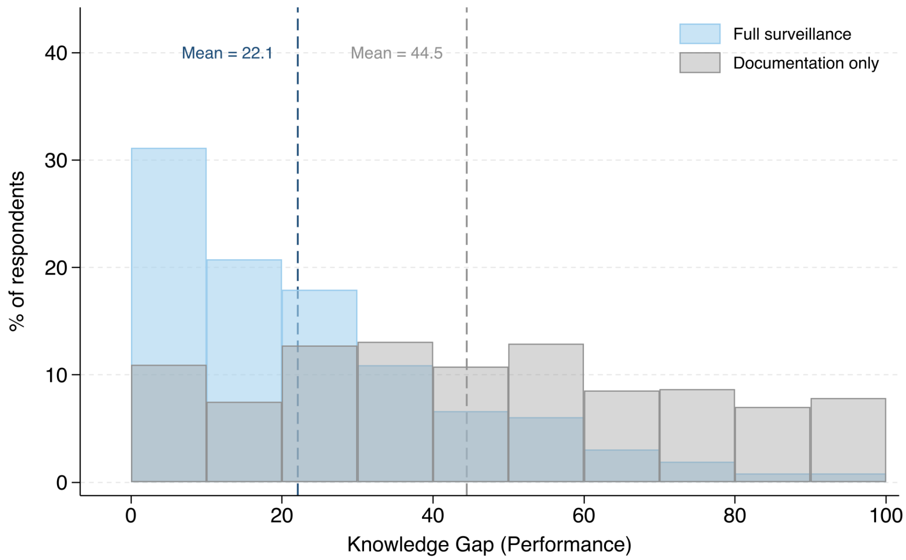
Figure 7: Knowledge Withholding and Knowledge Supply on the Knowledge Gap



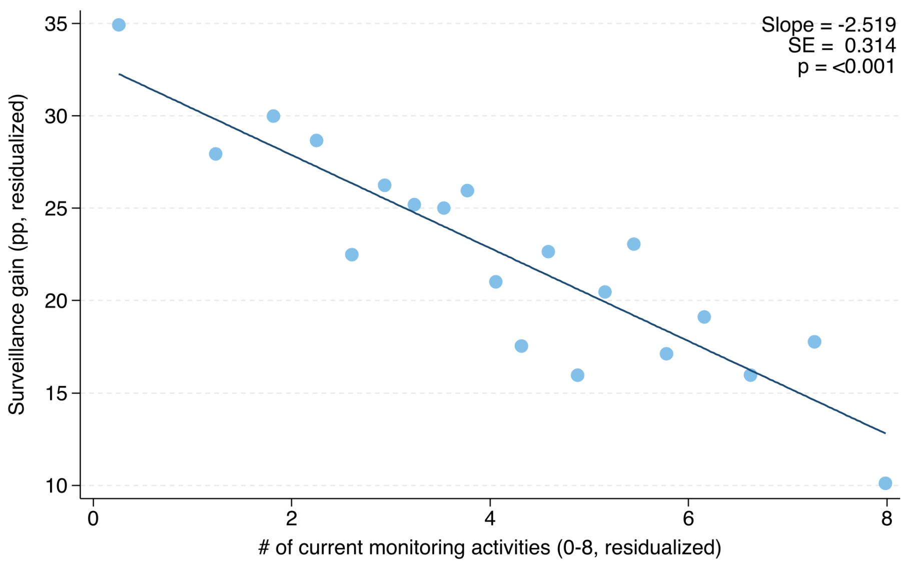
Notes: Bars show the mean Knowledge Gap (100 minus the respondent's rating of how well a similar replacement could perform their job), under three data scenarios. The grey bar is the baseline, in which the replacement learns only from the employer's existing documentation. In the withholding scenario (dark blue), the worker individually does everything they can to reduce the amount of data or documentation their employer is able to collect, while coworkers do not change their behavior. In the additional-supply scenario (light blue), the worker individually does everything they can to help the employer create additional data and documentation of their work activities and the thought process behind them, and the replacement learns from both the existing and the new documentation. Whiskers span ± 1 standard error of the mean.

Figure 8: Shrinking the Knowledge Gap via Surveillance

A. Distribution of the Knowledge Gap under Full Surveillance



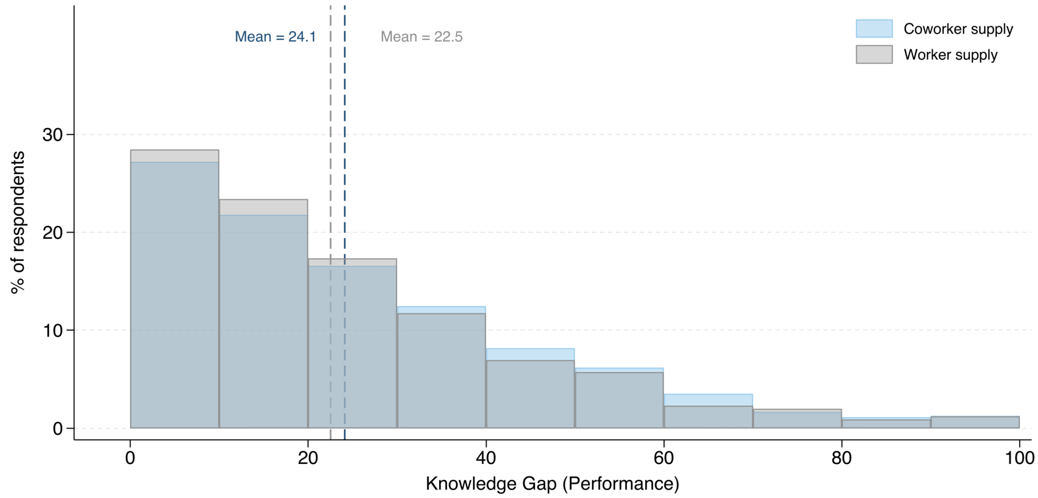
B. Surveillance Gain by Current Monitoring Intensity



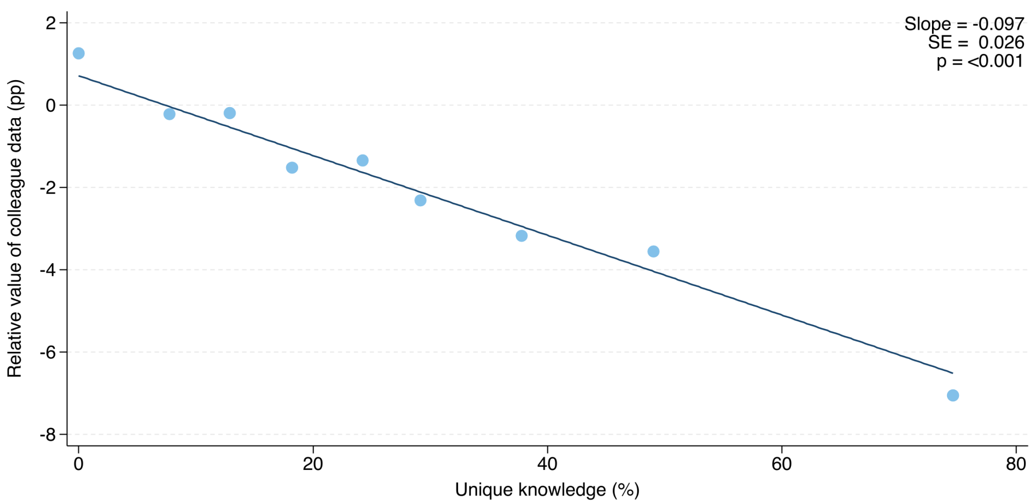
Notes: Panel A overlays two distributions of the Knowledge Gap (100 minus the respondent's rating of how well a similar replacement could perform their job). The grey distribution is the baseline, in which the replacement learns only from the employer's existing documentation; the blue distribution is the full-surveillance scenario, in which the replacement also learns from monitoring data covering everything the worker and their coworkers write, say, or do at work (excluding inner thoughts and decision-making). Dashed lines mark the two means. Panel B plots the surveillance gain, the reduction in the Knowledge Gap under full surveillance relative to documentation alone (in percentage points), against the firm's current monitoring intensity: the number of "*Yes*" responses across the eight data-collection channels in Appendix Figure A2 ("*Not sure*" responses are counted as not monitored). Panel B is a binned scatterplot (20 equal-sized bins) with both axes residualized on SOC 2-digit occupation and NAICS 2-digit industry fixed effects; the overlaid slope, standard error, and p-value come from the corresponding regression, with standard errors clustered by SOC 2-digit occupation group.

Figure 9: Shrinking the Knowledge Gap via Colleague Knowledge Supply

A. Distribution of the Knowledge Gap with Full Colleague Supply

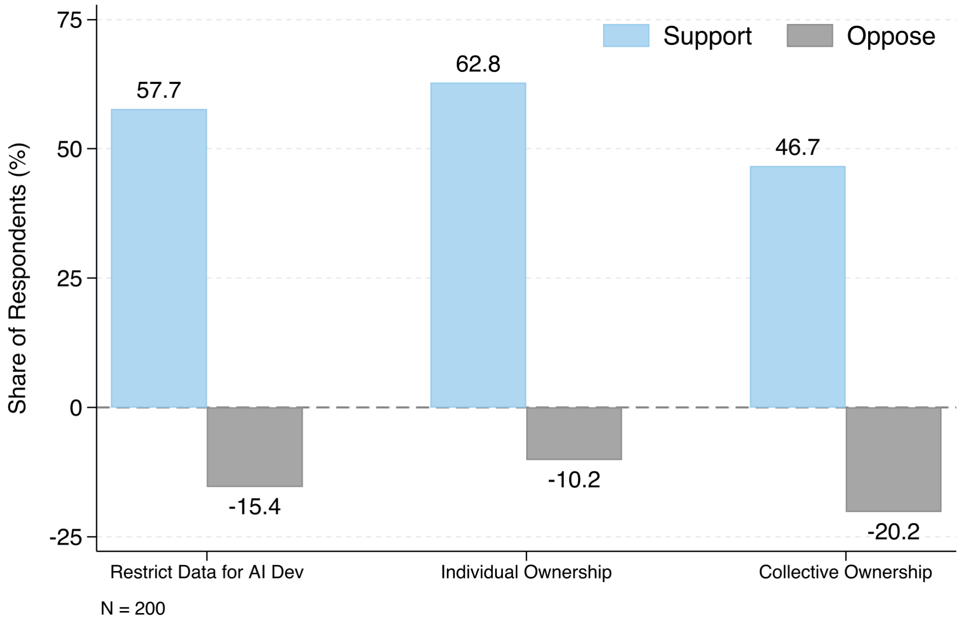


B. Relative Value of Colleague Data, by Worker's Unique Knowledge



Notes: Panel A overlays two distributions of the Knowledge Gap (100 minus the respondent's rating of how well a similar replacement could perform their job). In the grey distribution, the replacement learns from the employer's existing documentation plus all additional data and documentation the worker could individually supply; in the blue distribution, it learns from the existing documentation plus all additional data and documentation the worker's colleagues could collectively supply, with the worker's own behavior unchanged. Dashed lines mark the two means. Panel B plots the relative value of colleague data, the difference in rated replacement performance between the colleague-supply and worker-supply scenarios, against the worker's unique knowledge, the share of job knowledge captured neither by employer documentation nor by colleagues. Panel B is a binned scatterplot (10 equal-sized bins) with a linear fit. N = 2,158.

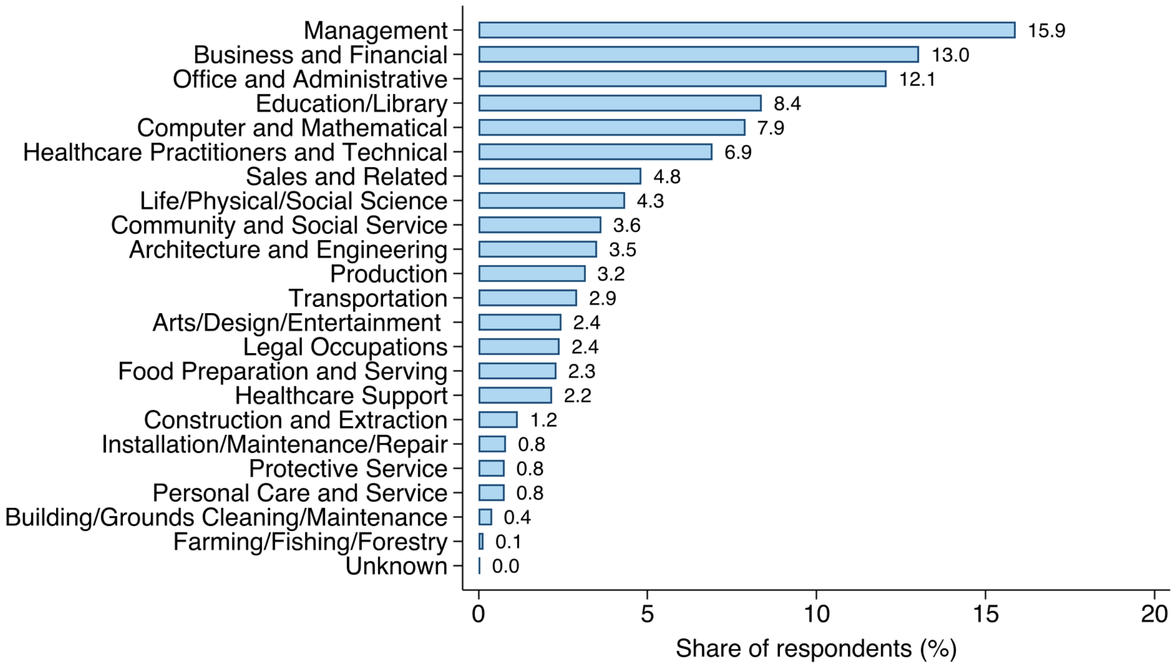
Figure 10: Policy Preferences



Notes: Bars show the share of respondents who support (blue, upward) or oppose (grey, downward) each of three policies governing work data, presented to respondents as experimental vignettes: a ban on using passively collected work data for AI development; individual ownership of work data, including the right to sell it; and collective ownership, under which workers jointly decide whether to sell their pooled data and how to divide the proceeds. For each policy, respondents were asked "*How would you feel if this policy were in place where you work?*"; support is the share answering "*Positive, I would like it*" and oppose is the share answering "*Negative, I would dislike it.*" "*Neutral*" responses are not shown. The policy questions were administered to a random subsample of respondents ($N = 200$). Section 3.5 describes the policy vignettes in further detail.

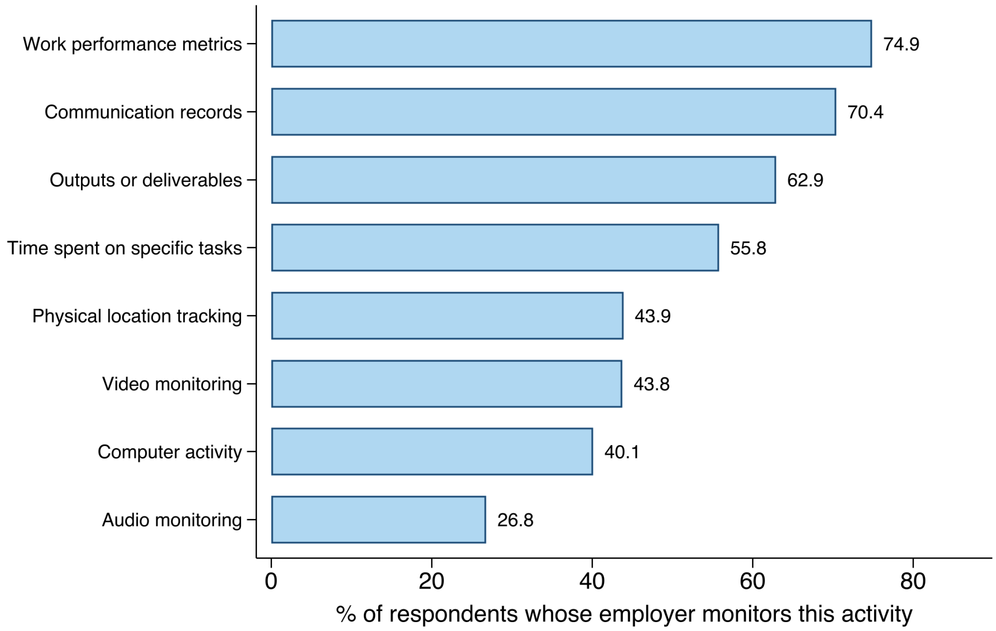
A Appendix Tables and Figures

Figure A1: Distribution of Respondents Across Occupations



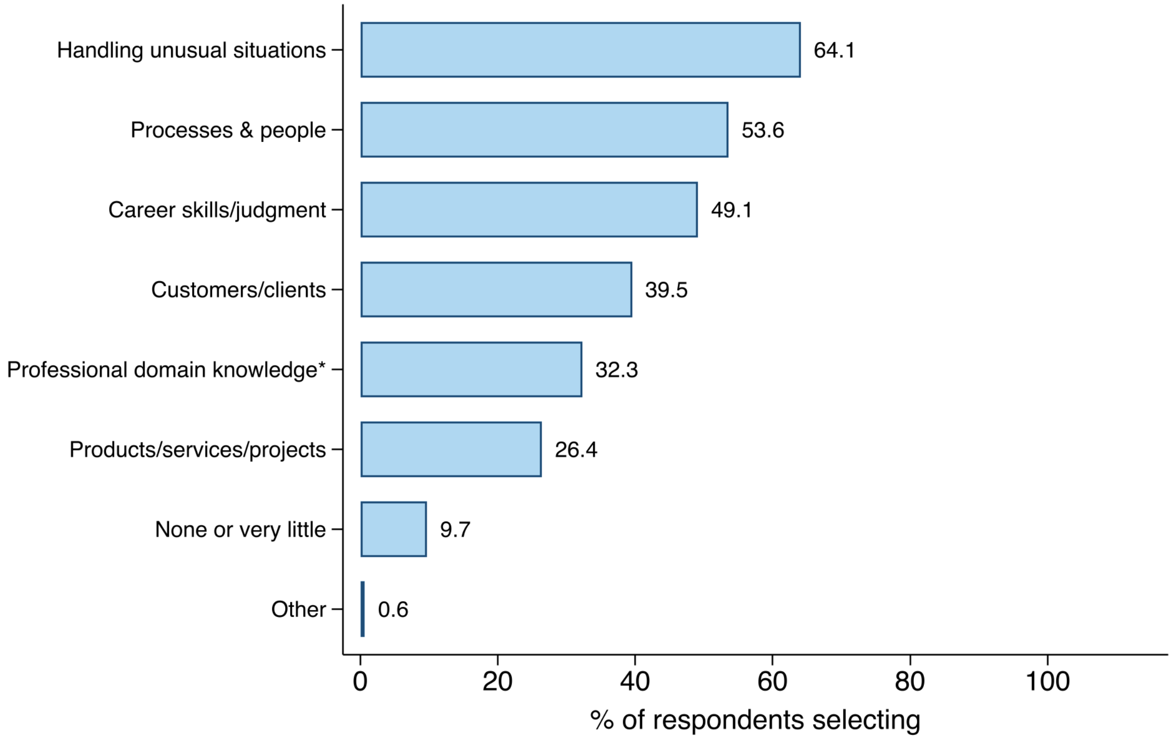
Notes: Share of the main analysis sample in each SOC 2-digit occupation group , sorted from largest to smallest. The sample comprises currently full-time employed U.S. workers recruited through Prolific and surveyed about their primary job. Recruitment was stratified across economic sectors, with sector-level quotas set so that the sample spans the full occupational distribution of U.S. full-time work rather than Prolific's default, which over-represents computer-based occupations. The figure reports unweighted shares; within strata, we construct post-stratification weights that match the weighted sample to the Census distribution of full-time workers on demographics (sex, age, and education) within major (SOC 2-digit) occupation groups, and unless noted otherwise, figures and tables report weighted statistics. The most common groups are management (15.9%), business and financial operations (13.0%), and office and administrative support (12.1%).

Figure A2: Prevalence of Workplace Monitoring, by Type



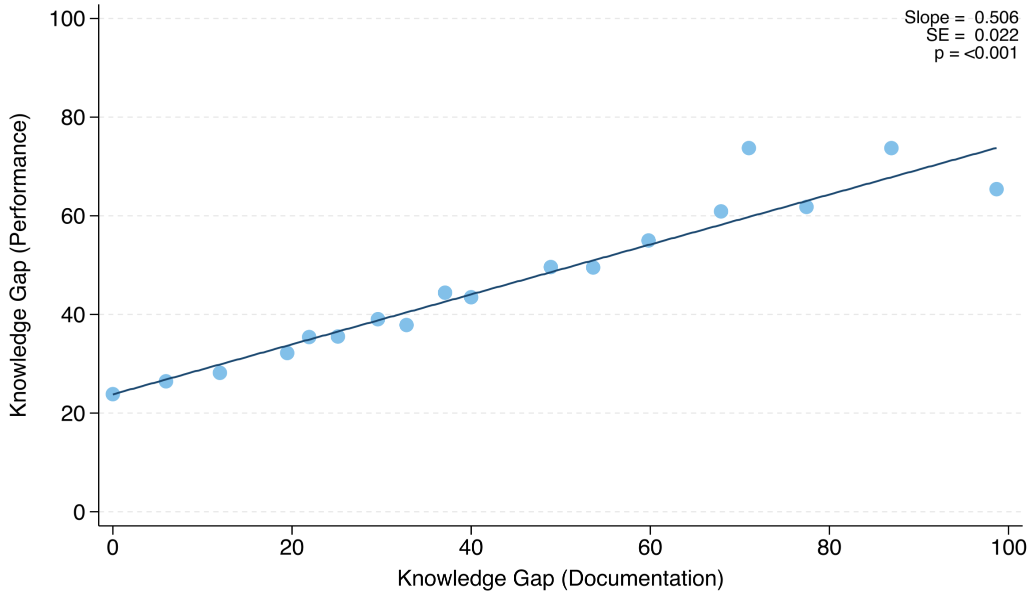
Notes: Share of respondents reporting that their employer monitors or collects data on each activity, from the multi-select question "*Which of the following activities does your employer monitor or collect data on?*" Response options for each activity were Yes / No / Not sure; bars report the share answering Yes, with No and Not sure responses included in the denominator. "*Not sure*" responses are counted as not monitored. The count of Yes responses across these eight channels is the surveillance-intensity measure (0–8) used in Figure 8 .

Figure A3: Work Knowledge Not Captured by Employer Documentation



Notes: Bars show the weighted share of respondents selecting each category in the multi-select question "*Reflecting on your answer above, what aspects of your work knowledge are important but not captured by the documentation your employer already has?*" Shares can sum to more than 100 percent. Categories, with the survey's examples: handling unusual situations (e.g., exceptions to rules, workarounds); processes and people (e.g., workflows, culture, how to work effectively with key individuals); general career skills or judgment (e.g., problem-solving, communication, experience); specific customers or clients (e.g., needs, history, relationships); professional domain knowledge (e.g., technical frameworks, regulations, and field-specific expertise); and specific products, services, or projects (e.g., specifications, market demand, known issues).

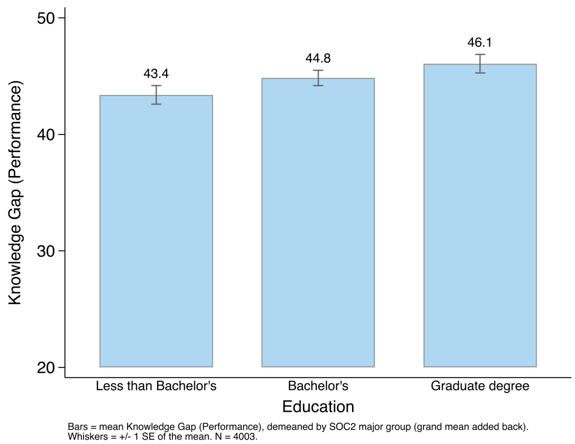
Figure A4: Correlation of Knowledge Gap (Performance) v. Knowledge Gap (Documentation)



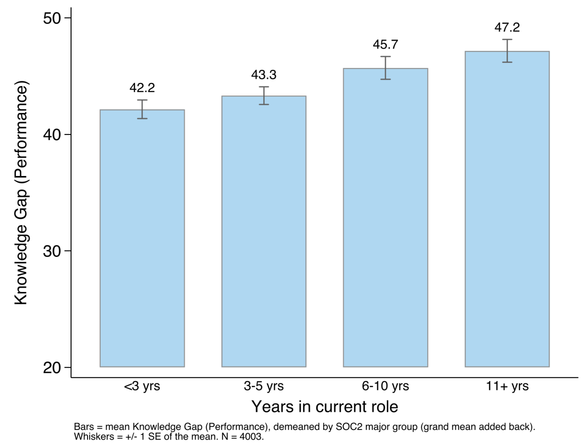
Notes: Binned scatterplot of the Knowledge Gap (Performance) against the Knowledge Gap (Documentation) at the individual level, both as defined in Figure 1: the documentation measure is the share of work knowledge not captured by the employer's documentation, and the performance measure is the shortfall in how well a replacement trained only on that documentation could perform the worker's job. Observations are grouped into 20 equal-sized bins of the documentation measure; each point plots the mean of both measures within a bin, and the line is the linear fit estimated on the underlying individual-level data using survey weights. The overlaid slope, standard error, and p-value come from that regression, with robust standard errors.

Figure A5: Within-Occupation Correlates of the Knowledge Gap

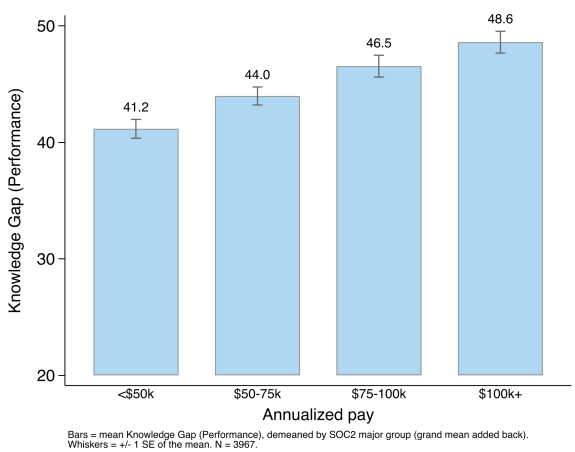
A. Education



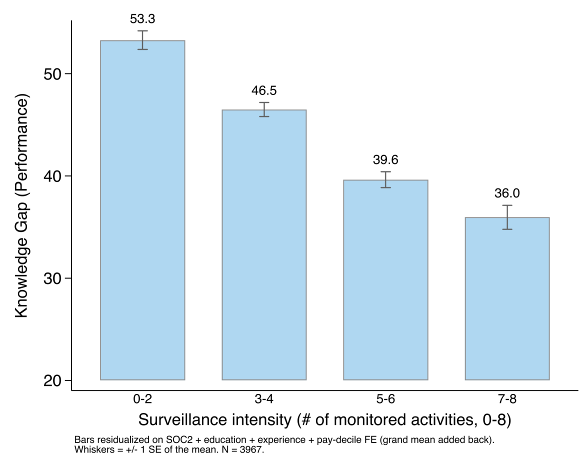
B. Experience in Current Role



C. Annualized Pay

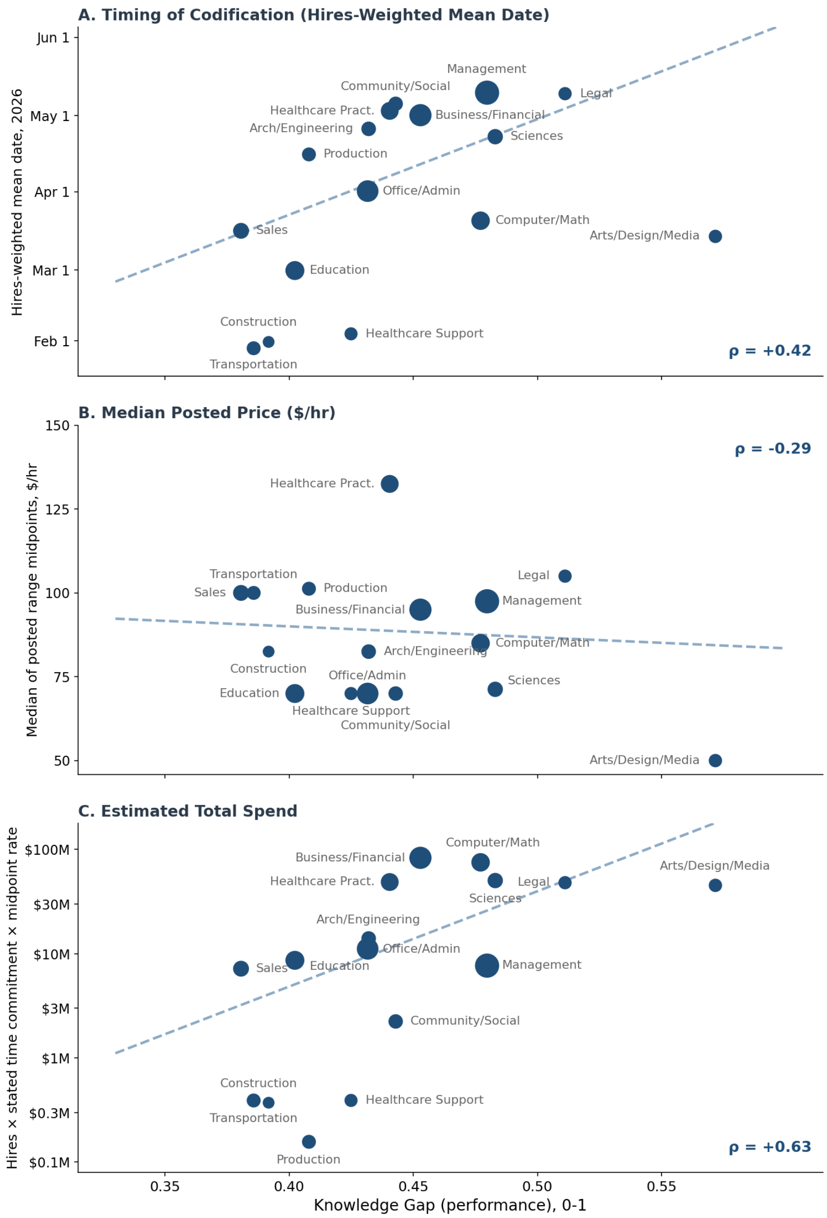


D. Surveillance Intensity



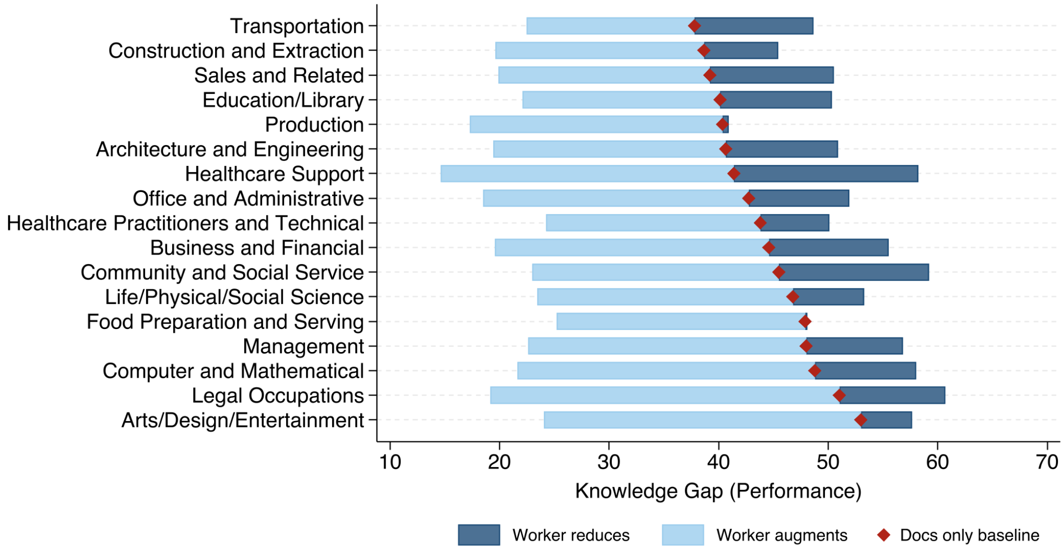
Notes: Each panel plots the mean Knowledge Gap (Performance), 100 minus the respondent's estimate of how well a replacement trained only on the employer's documentation and records would perform their job (as defined in Figure 1), by categories of a worker characteristic. The gap is demeaned by SOC 2-digit major group and the grand mean is added back; whiskers show ± 1 standard error of the cell mean. Annualized pay is annual base salary for salaried respondents and hourly wage $\times$ 2,080 hours for hourly respondents. Surveillance intensity is the number of the eight monitoring channels in Appendix Figure A2 that the respondent reports (binned). In Panel D, the gap is additionally residualized on fixed effects for education, experience in the current role, and deciles of annualized pay (grand mean added back), so the surveillance gradient is net of these worker characteristics.

Figure A6: A Market for Human Expertise: Mercor Hiring, Prices, and the Knowledge Gap



Notes: Data are daily observations of Mercor's public job board, January 20–July 13, 2026; each posting is classified from its title to a detailed SOC occupation (96% assigned at the SOC 2-digit level) and aggregated to the 2-digit major group. Each point is a SOC 2-digit occupation group (16 groups matched between the survey and Mercor hiring); marker size is proportional to survey respondents, dashed lines are OLS fits weighted by survey respondents, and ρ reports the correlation, computed on $\log_{10}$ of the plotted variable where its axis is log-scaled. The "Knowledge Gap (Performance)" on the horizontal axis measures the shortfall in how well a replacement could perform with access only to documentation, as defined in Figure 1. Panel A: the timing of codification, measured as the hires-weighted mean calendar date of the group's Mercor hiring. Panel B: the median across the group's hourly postings of the posted-range midpoint; project-priced postings without an hourly rate are excluded. Panel C: estimated total spend = each posting's hires × its stated time commitment × its posted midpoint rate, summed over the group's postings, on a log scale. Healthcare Support, Construction, Production, and Transportation each have fewer than 100 Mercor hires over the panel.

Figure A7: Worker Knowledge Supply and Withholding Capability by Occupation



Notes: Each row is a SOC 2-digit occupation, limited to groups with at least 40 respondents. The red diamond marks the mean Knowledge Gap (Performance), 100 minus self-reported replacement performance (the shortfall in how completely a replacement could perform the worker's job, on a 0–100 scale), under employer documentation alone. The light blue segment extends to the mean gap when the worker additionally *supplies* their own data and documentation; the dark blue segment extends to the mean gap when the worker instead *withholds* / reduces data collection. The width of the band measures how far an individual worker can widen or narrow their own perceived knowledge gap relative to the documentation-only baseline.

Table A1: Comparison of Survey and CPS Sample Characteristics

	Survey	CPS	Difference
Age	37.07	42.15	-5.08
	(10.36)	(13.23)	
Male	0.403	0.546	-0.142
White	0.690	0.750	-0.060
Black	0.133	0.133	0.000
Asian	0.069	0.078	-0.010
Private Sector	0.831	0.834	-0.003
Unionized	0.153	0.110	0.043
Salaried	0.533	0.495	0.038
Hourly	0.467	0.505	-0.038
Annual Salary ($)	89,307.00	106,729.88	-17,422.88
	(45,787.26)	(103,062.49)	
Hourly Wage ($)	26.64	35.46	-8.83
	(25.64)	(19.76)	
N	4,043	50,313	

Notes: This table compares summary statistics between our survey sample and a cross-section of U.S. full-time workers, drawn from the 2025 Annual Social and Economic Supplement (ASEC) of the Current Population Survey (CPS). The CPS sample is restricted to employed, full-time wage and salary workers aged 18 and older. "Private Sector" comprises for-profit and non-profit employers, as opposed to government. Self-reported survey earnings are screened for unit-entry errors: hourly wages above $1,000 and annual salaries below $15,000 are set to missing. Standard deviations for continuous variables are shown in parentheses. CPS demographic statistics are weighted with ASEC person weights; union status, the salaried/hourly classification, hourly wage, and annual salary are computed over the CPS earner (Outgoing Rotation Group) subset, approximately one-quarter of the sample, and are weighted with the ORG earnings weight. CPS annual salary reflects prior-year wage income.

B Proofs

B.1 Proof of Theorem 4.9

Proof. The worker payoffs (7) from time-1 contributions induced by each technology are

$$w_1^j - c^j(k_1^j) + \psi\gamma \left[\theta_j - \alpha^j\left(k_1^{\overline{J}_1 \cup \{j\}}\right)\right]_+ , \tag{35}$$

$$w_1^j - c^j(k_1^j) + \psi\gamma \left[\theta_j - \tilde{\alpha}^j\left(k_1^{\overline{J}_1 \cup \{j\}}\right)\right]_+ . \tag{36}$$

By the same argument as in Lemma 4.4, these define a pair of supermodular games. Let us index these games by parameter τ, with $\tau = 0$ corresponding to $\left(\alpha^j\right)_{j\in J}$ and $\tau = 1$ corresponding to $\left(\tilde{\alpha}^j\right)_{j\in J}$. By (25), (26), and Lemma 4.3, it follows that each worker j's utility has increasing differences in (k_1^j, τ). Thus, the comparison of extremal equilibria follows by Theorem 6 of Milgrom and Roberts (1990).

Finally, observe that worker j's time-2 wage under the $\overline{k}_1$ and α is

$$\lfloor \theta_j - \alpha^j \left(\overline{k}_1\right) \rfloor_+ \geqslant \left\lfloor \theta_j - \alpha^j \left(\overline{\tilde{k}}_1\right) \right\rfloor_+ \geqslant \left\lfloor \theta_j - \tilde{\alpha}^j \left(\overline{\tilde{k}}_1\right) \right\rfloor_+, \tag{37}$$

where the first inequality is by α non-decreasing and the second inequality by (25). The right-hand side of (37) is worker j's time-2 wage under the $\overline{\tilde{k}}_1$ and $\tilde{\alpha}$. The same argument holds comparing wages under $\tilde{\underline{k}}_1$ and $\underline{k}_1$. □

B.2 Proof of Theorem 4.12

Proof. By (26) we have

$$\alpha^l(\theta) - \alpha^l(0, \theta^{-j}) \geqslant \tilde{\alpha}^l(\theta) - \tilde{\alpha}^l(0, \theta^{-j}) \text{ for any workers } j \text{ and } l. \tag{38}$$

By (31) and (38), we have

$$-\alpha^l(0, \theta^{-j}) \geqslant -\tilde{\alpha}^l(0, \theta^{-j}) \text{ for any workers } j \text{ and } l. \tag{39}$$

Worker j's payoff under $\left(\alpha^l\right)_{l \in J}$ is

$$\begin{aligned} &\gamma\theta^j + \psi\gamma \left(\max\left\{\theta^j, \alpha^j(\theta)\right\} - \alpha^j\left(0, \theta^{-j}\right) + \sum_{l \neq j} \left(\max\left\{\theta^l, \alpha^l(\theta)\right\} - \max\left\{\theta^l, \alpha^l\left(0, \theta^{-j}\right)\right\} \right) \right) \\ &\geqslant \gamma\theta^j + \psi\gamma \left(\max\left\{\theta^j, \tilde{\alpha}^j(\theta)\right\} - \tilde{\alpha}^j\left(0, \theta^{-j}\right) + \sum_{l \neq j} \left(\max\left\{\theta^l, \tilde{\alpha}^l(\theta)\right\} - \max\left\{\theta^l, \tilde{\alpha}^l\left(0, \theta^{-j}\right)\right\} \right) \right), \end{aligned} \tag{40}$$

where the inequality is by (31) and (39). The right-hand side is equal to worker j's payoff under $\left(\tilde{\alpha}^l\right)_{l \in J}$ □

B.3 Proof of Theorem 4.13

Proof. We are comparing full-contribution equilibria, so workers incur no withholding costs. Thus, it suffices to show that workers' wages are lower under individual data ownership than under no surveillance.

There are at least two workers, they have identical maximum contributions, and their contributions are perfect substitutes, so expression (27) for pairwise surplus at time 2 under individual ownership reduces to

$$\max\left\{\theta^j, \alpha\left(k_1^J\right)\right\} - \alpha\left(k_1^{J\setminus\{j\}}\right) = \left\lfloor\theta^j - f(\theta^j)\right\rfloor_+, \tag{41}$$

which is strictly less than θ^j by $f(0) = 0$, $\theta^j > 0$, and f increasing. Thus wages at time 2 under individual data ownership are strictly lower than $\gamma\theta^j$ by $\gamma \in (0, 1]$, which is the wage under no surveillance.

Next we consider time-1 wages. There are at least three workers, so deviating to not hire a single worker at time-1 has no effect on time-2 output or on time-2 wages. Thus, the pairwise surplus of hiring worker j at time-1 is equal to θ^j, and time-1 wages are the same under individual ownership and under no surveillance.

We have established that individual ownership results in the same wages at time 1 and strictly lower wages at time 2, which by $\psi > 0$ implies that workers are strictly worse off. □

B.4 Proof of Theorem 4.16

Proof. Observe that even under costly contributions, the pairwise surplus from employing an additional worker at time 2 is at least zero, so full employment at time 2 is consistent with equilibrium. Given full employment at time 2, worker j's payoff is still as in (28). By $\left|K^j\right|$ finite and the existence of Nash equilibrium in finite games, for any employed set $\overline{J}_1$, there exists a (possibly mixed) profile of time-1 contributions in which each worker's choice of k_1^j maximizes their expected payoff (that is, the expectation of (28)) given $\overline{J}_1$ and the other workers' mixtures over contributions.

It remains to show that full employment at time 1 is consistent with the equilibrium we are constructing. Observe that, under full employment at time 1, the pairwise surplus of the firm and worker j at time 1 (their time 1 payoffs plus their continuation payoffs) is

$$\underbrace{k_1^j - c^j(k_1^j)}_{\text{time 1 output and cost}} + \underbrace{\psi \max\left\{\theta^j, \alpha^j\left(k_1^{J_1}\right) - \alpha^j\left(k_1^{J_1 \setminus \{j\}}\right)\right\}}_{\text{change in continuation payoff for role } j}$$
$$+ \underbrace{\psi \sum_{l \neq j} \left((1-\gamma)\left\lfloor \max\left\{\alpha^l\left(k_1^{J_1}\right), \theta^l\right\} - \alpha^l\left(k_1^{J_1 \setminus \{l\}}\right)\right\rfloor_+ + \alpha^l\left(k_1^{J_1 \setminus \{l\}}\right)\right)}_{\text{time-2 profit from other roles with } j\text{'s data}}$$
$$- \underbrace{\psi \sum_{l \neq j} \left((1-\gamma)\left\lfloor \max\left\{\alpha^l\left(k_1^{J_1 \setminus \{j\}}\right), \theta^l\right\} - \alpha^l\left(k_1^{J_1 \setminus \{j,l\}}\right)\right\rfloor_+ + \alpha^l\left(k_1^{J_1 \setminus \{j,l\}}\right)\right)}_{\text{time-2 profit from other roles without } j\text{'s data}}. \tag{42}$$

The above expression is at least zero, by (34) and α^j, α^l non-decreasing, which completes the proof. $\square$

B.5 Proof of Theorem 4.17

Proof. Let κ_1^{-j} be the random variable representing other workers' time-1 contributions under some equilibrium. Observe that for k_1^j played with positive probability, we have

$$k_1^j \in \arg\max_{\hat{k}_1^j} E\left[w_1^j - c^j(\hat{k}_1^j) + \psi\gamma\lambda_j\left(\hat{k}_1^j, \kappa_1^{-j}, \overline{J}_1\right)\right], \tag{43}$$

where λ_j is as defined in (27). Moreover we have

$$\left\{\theta^j\right\} = \arg\max_{\hat{k}_1^j} E\left[w_1^j - c^j(k_1^j)\right], \tag{44}$$

because θ^j is the unique minimizer of c^j. Since $E\left[\lambda_j\left(\hat{k}_1^j, \kappa_1^{-j}, \overline{J}_1\right)\right]$ is non-decreasing in $\hat{k}_1^j$, the result follows by Topkis's theorem. $\square$

B.6 Proof of Theorem 4.18

Proof. Let $\mathbf{k}_1$ be the random variable representing the time-1 contribution vector in an equilibrium with collective ownership. For any worker j, they cannot profitably deviate to contribute θ^j at time

1, so for any k_1^j in the support of $\mathbf{k}_1$ we have

$$E\left[-c^j(k_1^j)+\gamma\psi\frac{\sum_l \max\left\{\theta^l,\alpha\left(k_1^j,\mathbf{k}_1^{-j}\right)\right\}}{|J|}\right]\geqslant E\left[\gamma\psi\frac{\sum_l \max\left\{\theta^l,\alpha\left(\theta^j,\mathbf{k}_1^{-j}\right)\right\}}{|J|}\right]\geqslant\gamma\psi\frac{\sum_l\theta^l}{|J|}. \tag{45}$$

Summing (45) across j and by $\gamma\in[0,1]$ we have

$$\sum_j E\left[-c^j(\mathbf{k}_1^j)+\psi\max\left\{\theta^l,\alpha\left(\mathbf{k}_1\right)\right\}\right]\geqslant\sum_j\psi\theta^j. \tag{46}$$

By the same argument as in Section B.5, if k_1^j is in the support of $\mathbf{k}_1$, then we have $k_1^j\geqslant\theta^j$. Thus by (46) we have

$$\sum_j E\left[\mathbf{k}_1^j-c^j(k_1^j)+\psi\max\left\{\theta^l,\alpha\left(\mathbf{k}_1\right)\right\}\right]\geqslant\sum_j\left(\theta^j+\psi\theta^j\right). \tag{47}$$

The left-hand side of (47) is the expected total surplus under collective ownership and the right-hand side is the total surplus in the no-AI case. □